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NPS/003/014 – Technical Specification for Ground Mounted 11 & 20kV Distribution Switchgear

1. Purpose

The purpose of this document is to specify the technical requirements for 11 and 20kV switchgear for use at ground mounted substations operating on the CE Electric UK network.

This specification seeks to be functional and to clarify, or vary, existing national specifications and requirements only where necessary.

This document supersedes the following documents, all copies of which shall be destroyed.

Ref	Version	Title
NPS/003/014	1.0	Technical Specification for 11 & 20kV Distribution Switchgear
DSS/011/013		11kV Distribution Switchgear
NE MS 11	Issue 6	Specification for Metal Enclosed Ring Main Equipment
SDC/PRC/S/209		11kV Distribution Switchgear

2. Scope

This specification covers: cable connected or transformer mounted, extensible and non-extensible ring main units, extensible switchgear and metering units, having a rated voltage of 12 or 24kV.

This specification does not cover switchgear rated and intended for auto-reclosing duties.

3. Technical Requirements

3.1 Technical Specification

3.1.1 Overall

The switchgear shall comply fully with, and meet all the requirements of, the current version of “ENATS 41-36 - Distribution Switchgear For Service Up To 36kV (Cable And Overhead Conductor Connected)”, except where varied by this specification.

The switchgear shall, preferably, have been assessed by the Energy Networks Association and granted a Notice of Conformance against ENATS 41-36.

3.1.2 Environment and Suitability for Indoor or Outdoor use

Switchgear is required for use in two general categories:

- 3.1.2.a Indoor; typically a brick built, GRP, or sheet steel housing. This will provide protection from the direct effects of most elements of the weather, but is not usually temperature maintained, or

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climate controlled. The housing will usually have multiple, large louvered air vents and is unlikely to prevent dust, high humidity air or salt fog entering the housing.

3.1.2.b Outdoor; Open to the direct effects of all UK weather elements and climatic effects.

Note :- HV metering units will not be used outdoors.

3.1.3 Remote Control

All circuits on all switchgear types shall be supplied with the capability of being remotely controlled and operated. The functionality and operational safety requirements of the remote control option shall fully comply with the CE Electric UK technical specification NPS/003/017.

Where switchgear is not fitted with remote control actuators at the time of order; facilities shall be provided for the fitting of these facilities retrospectively. Where the provision of wiring, connection points, or auxiliary switches to facilitate remote control can only be installed during manufacture then these shall be incorporated as standard. The retrofitting of any actuator and all associated components: shall be non-intrusive, not require material modification of the switchgear and not require the switchgear to be taken out of normal service condition or require the operational position of the switchgear to be changed.

To enable an effective remote control facility to be realised, all switchgear shall have the capability for the incorporation of fault sensing CTs. See clause 3.7 of this specification for further details.

3.2 Protection

The circuit breaker shall be provided with a protection system that provides a minimum of 2 pole overcurrent and single pole earth fault protection. This shall be provided utilising either time fuse links, or a self powered relay.. If the standard protection arrangement utilises time limit fuses then an option for a self powered relay shall be provided.

630Amp circuit breakers shall be provided with a self powered, two pole overcurrent and earth fault time delayed (IDMT) multi characteristic relay. The selectable protection characteristics shall include: IEC IDMTL normal inverse (NI), very inverse (VI), extremely inverse (EI) curves and definite time lag (DTL) protection.

3.2.1 Relays

All relays shall have been successfully assessed by the Protection Assessment Panel of the Energy Networks Association, against ENA-TS-48-5 and ENA-TS 48-6-6.or shall be able to demonstrate compliance with the self certification schedules detailed in Appendix 5J & 5K of this specification.

Additionally all relays shall comply fully with the requirements of IEC 60255 and shall, as a minimum, provide the following features:

Self powered operation, without internal battery.

Easily re-settable, without the use of special tools, or IT equipment.

Unambiguous visual indication of operation. The relay and this indication shall be capable of remaining indefinitely in the operated state.

Protection settings/configuration shall be directly selectable, via the relay front panel, without the use of electronic interface equipment.

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3.3 Metering

The specification for metering is dictated by the relevant meter owner and/operator for the individual site. Circuits with Embedded Generation shall have an option for a voltage transformer to measure residual voltage and for category 5P10 CTs with minimum 15VA capacity. Some typical examples of metering specifications are given in Appendix 1.

Switchgear that will be used to give metered HV supplies to customers shall be provided with facilities for accepting both 110V AC and/or 30V DC remote emergency tripping signals.

3.4 Testing

CE Electric UK policies and procedures require switchgear to be over-voltage tested in accordance with the NEDL-YEDL Operational Practice Manual. The test voltages appropriate to this equipment are specified in Table WE.1 in section WE TESTING; a copy of this table is included as Appendix 2.

The switchgear shall be designed to allow the test voltages to be applied to associated cables and/or plant whilst these are connected to the switchgear, without any degradation in performance to the switchgear.

Where VTs are installed for metering or protection purposes it shall be possible to easily isolate the VTs from the switchgear primary circuit, to allow testing of the switchgear, any connected cables and/or plant.

Access to the protection CT secondary wiring terminals shall be provided to allow secondary injection testing to be carried out.

CE Electric UK require manufacturers to carry out a number of additional pre-commissioning checks on the equipment prior to its final despatch to CE Electric UK. These tests are detailed in Appendix 2.

3.5 Resistance to Physical Interference/Damage

Conductors, such as separable connectors, that are not surrounded by a continuous metallic earth screen shall be enclosed in an earthed metal casing that:

- When installed on the switchgear provides physical protection, tested to 5J impact at the most vulnerable point on all exposed sides and at the vulnerable point on each exposed edge.
- Provides a minimum IP rating of IP21B in accordance with IEC 60529. For the purposes of this test the unscreened insulation is to be classed as a hazardous part that requires 20mm clearance from the end of the IP21B test probe.

3.6 Variances from ENATS 41-36

This Section is numbered to reflect the clause numbers of ENATS 41-36 to which the variances apply.

1.4 Ratings

1.4.2.1 Table 1.1a Normal Rated Insulation Levels;

For 12kV switchgear the higher values of Rated Lightning Impulse Withstand Voltage (Up) shall be applied; i.e. 95kV/110kV (peak value) shall be the minimum.

1.4.4.1 Table 1.3 Rated Normal Current; busbars shall have a current rating of at least 630A.

1.4.4.1 Table 1.3 Rated Short-Time Withstand Current:

12kV rated units shall have a current rating of at least 20kA.

24kV rated units shall have a current rating of at least 16kA.

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1.4.8 Rated Supply Voltage of Closing and Opening Devices and of Auxiliary and Control Circuits

Remote control facilities will be rated to operate with 24V DC supplies.

All switchgear utilised for metered HV supplies shall be provided with facilities for accepting 110V AC and/or 30V DC remote emergency tripping signals

1.5 Design and Construction

1.5.1 Requirements for Liquids

Oil filled switchgear will not normally be acceptable.

1.5.2 Requirements for Gases in Switchgear and Controlgear

The manufacturer shall state whether 'topping up' can be carried out with the switchgear live and shall provide instructions for carrying out this operation.

1.5.8 Operation of Releases

All switchgear utilised for metered HV supplies shall be provided with facilities for accepting remote signals for emergency tripping.

1.5.9 Low and High Pressure Interlocking and Monitoring devices

A gauge or indicator (option a) shall always be provided for each gas pressure system.

1.5.13.2 Protection against Ingress of Water.

Switchgear, including metering units categorised as 'Indoor' shall be certified to at least IPX1, and preferably to IPX2, to protect against damage from water ingress into the room, housing or enclosure.

1.5.15 Gas and Vacuum Tightness

The leakage rate shall not exceed 0.5% per year at 20°C

1.5.101 Internal Fault

Options for avoiding, or restricting, the venting of arc products into the substation building shall be provided; e.g. ducting to divert arc products out of the building, arc quenchers/filters, etc.

1.5.102.103 Transformer Mounting Arrangement

12 kV rated switchgear might be required to be directly connected to the HV flange of a 'Cable Connected Transformer' as specified in ENATS 35-1. The transformer flange height and facing will be the same as for a 'Unit Substation Transformer'. A busbar extension option is required for use in situations where the transformer radiator arrangement restricts the space available for the switchgear.

1.5.103.1.101 Cable Compartments; the use of separable connectors is the preferred method of terminating HV cable onto the switchgear. Where this is available the following bushing interfaces shall be provided:-

Feeder Cable terminations

Dead Break plug-in cable terminations with bolted contact (M16) as interface type "C" according to EN 50181/Din EN 50181 rated for 400/630A.

Transformer Cable terminations

Dead Break plug-in cable terminations with bolted contact (M16) as interface type "C" according to EN 50181/Din EN 50181 rated for 200A

All units are to be supplied complete with cable support clamps capable of accommodating the range of cable sizes detailed in the CE Electric UK network product specification NPS/002/020

1.5.201.3 Voltage Checks and phase identification

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A VPIS, or VDS, system for phase comparison shall be provided on all switchgear primary circuits.

1.5.201.5 Test Devices

Particular attention must be paid to ensure that the insertion, removal or replacement of any components (e.g. test bushings, separable connector blanking plugs, etc) required as part of the procedures to facilitate testing activities can be carried out in an ergonomically acceptable manner. Manufacturers shall provide detailed instructions on how this can be achieved.

1.10 Rules for Transport, Storage, Installation, Operation, Maintenance and Disposal

The installation, testing, commissioning, operating, maintaining and removal procedures for the switchgear shall take into account the requirements of UK safety legislation and requirements, including legislation governing working at heights, and shall be able to demonstrate compliance with all of these.

The switchgear must be able to be installed in existing substations or third party buildings which preclude the use of mechanical lifting devices. It shall be possible to manually move and locate the switchgear in a safe and ergonomically acceptable manner.

2.4 Ratings

All Circuit Breakers shall be rated for 'line-charging breaking current'

4 Switch Fuse Combinations

The characteristics of the feeder protection at primary substations effectively preclude the addition of switch fuse combinations onto the CE Electric UK distribution networks.

10.6 Fault Passage Indication

10.6.1 Fault Passage Indication

Ring Main Units and Extensible Switchgear Units shall have integral CTs able to detect earth fault, or phase fault current. These CTs shall: fully comply with CE Electric UK network product specification NPS/003/013, provide an output to operate a fault passage indicator relay that complies with ENATS 48-2 types A & B and be terminated in a termination cubicle that is easily accessible from the front of the switchgear.

All Ring main Units shall have these CTs on the circuit associated with the left hand side feeder, as viewed from the normal operating position of the switchgear; but all circuits on all switchgear types shall have the option for provision of these CTs.

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4. References

4.1 External Documentation

Reference	Title
ENATS 41-36	Energy Networks Association; Distribution Switchgear For Service Up To 36kV (Cable And Overhead Connected)
ENATS 35-1	Energy Networks Association; Distribution Transformers (from 16kVA to 2000kVA)
ENA-TS-48-5	Energy Networks Association; Environmental Test Requirements for Protection Relays and Systems
ENA-TS-48-6-6	Energy Networks Association; Functional Test Requirements – Overcurrent and Earth Fault Protection Equipment
ENA TS 50-19	Energy Networks Association. “Standard Numbering for Small Wiring”
ENA TS 50-18	Energy Networks Association. “Application of Ancillary Electrical Equipment”
IEC 60529	Degrees of protection provided by enclosures (IP code)
IEC62271-1	High-voltage switchgear and controlgear - Part 1: Common specifications
IEC 60694	Common specifications for high-voltage switchgear and controlgear standards
IEC 60255 series	Measuring relays and protection equipment

4.2 Internal documentation

Reference	Title
NPS/002/020	Technical Specification for 11 & 20kV Cables
NPS/003/013	Technical Specification for Fault Passage Indicators
NPS/003/017	Technical Specification for a Ground Mounted Distribution Switchgear Remote Control System

5. Definitions

Term	Definition
NONE	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant & Switchgear Standards Manager	D Blackledge	03/09/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System strategy	M Nicholson	07/09/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	09/09/2009

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	CDS Administrator	S Johnson	05/10/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	03/09/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	09/09/2009

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Appendix 1 –Variants Required for use by CE Electric UK

Typical Unit Types Required by CE Electric UK

Ring Main Unit – Non- Extensible - Close Coupled. With tee circuit directly connectable, or trunk connectable, to ENATS 35-1 transformer HV flange and also cable connectable by the addition of a cable box onto the tee off flange. Minimum of 3 devices, of which at least one must be a circuit breaker
Ring Main Unit – Non- Extensible - Cable Connected. Cable connected on all circuits
Ring Main Unit – Extensible - Close Coupled. With tee circuit directly connectable, or trunk connectable, to ENATS 35-1 transformer HV flange and also cable connectable by the addition of a cable box onto the tee off flange. Minimum of 3 devices, of which at least one must be a circuit breaker
Ring Main Unit – Extensible - Cable Connected. Cable connected on all circuits
Circuit Breaker - Extensible
Circuit Breaker - Non-Extensible- Close Coupled. With earthing facilities on load side, in addition to fully rated earthing facilities on the supply side.
Circuit Breaker - Non-Extensible- Cable Connected. With earthing facilities on load side, in addition to fully rated earthing facilities on the supply side.
Switch (switch disconnecter) - Extensible
Busbar Earth Switch - Extensible
Cable box for RMU tee off circuit.
Busbar cable box for any/all Extensible switchgear units.
Trunk connection unit to allow busbar connection of switchgear unit to ENATS 35-1 Transformer, where direct connection is not possible due to physical restrictions (e.g. transformer radiator arrangement).
Insulated end cap , fully rated, for all/any extensible busbars.
Metering Units - as detailed in the Metering Unit section below.

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Metering Units Required by CE Electric UK

Overview:

The metering unit shall be in accordance with ENATS 41-36.

Requirements of metering units are dictated by agreement between the meter owners/operators/installers and may vary from site to site. However, the following rules and examples are, in the absence of any other directions, typical requirements for use on CE Electric UK networks.

Metering units are generally:

- Directly coupled to Ring Main Units,
- Directly coupled to extensible switchgear,
- Free standing independent units.

Due to multiple parties being involved in the installation, commissioning and operation of the metering unit, it is of particular importance that the secondary and auxiliary wiring in the metering unit and associated equipment is fully compliant with both ENATS 50-19 Energy Networks Association. "Standard Numbering for Small Wiring" and ENATS 50-18 Energy Networks Association. "Application of Ancillary Electrical Equipment"

To allow termination of the cables associated with metering; a metal enclosed multicore cable box shall be provided, complete with cable glands complying with BS6121: Type E1W or equivalent.

Because multiple parties are involved in the installation, commissioning and operation of the metering; access to this cable box it is strongly preferred that access to this cable box is independent of access into any switchgear or controlgear compartments.

Access to the metering terminals shall not require the use of tools.

VTs shall be at least class 1.0, have a minimum output of 100VA per phase and have a rated voltage factor of 1.2 continuous and 1.9 for 30 seconds.

VTs shall, preferably, be connected to the primary conductors through readily removable fuse links with current limiting characteristics to BS2692, or equivalent.

CTs shall have a minimum output of 7.5VA and an accuracy class of at least 0.5S.

Common return, or feed, conductors for two or more metering secondary circuits are not permitted.

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Schedule of Typical Metering Units

ITEM NO	DESCRIPTION
1	2 x 100/50/5 A current transformers for metering, class xxx 1 x voltage transformer, ratio 11,000/110 V, class xxx, burden yyy
2	2 x 50/25/5 A current transformers for metering 1 x voltage transformer, ratio 11,000/110 V, complete with HV and LV fuses
All	Alternative voltage transformers might include: 11,000-6,600/110 V in place of 11,000/110 V VT 11,000-5,500/110V in place of 11,000/110 V VT

Schedule of Typical Metering Unit for a Site that Exports onto CE Electric UK's Network

ITEM NO	DESCRIPTION
1	<u>11kV Metering Unit</u> 2 x 100/50/5 A, 15VA 5P10 CTs 1 x voltage transformer, ratio 11,000/110 V, class xxx, burden yyy
2	<u>20kV Metering Unit</u> 2 x 100/50/5 A, 15VA 5P10 CTs 1 x voltage transformer, ratio 20,000/110 V, class xxx, burden yyy

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Appendix 2 – Testing

(a) Test Voltages Applied By CE Electric UK.

As defined in 'WE TESTING' section of CE Electric UK's Operational Practice Manual

Table WE.1 Test Voltages (kV)

Column 1	2	3	4	5	6	7
Working Voltage	DC 15 mins	AC 1 min	DC 1 min	AC 15 mins	DC 1 min	AC 1 min
11kV	20	16	5		28	20
11/20kV Cable	28		5			
20kV	37	30	5		55	50

Notes:

Column 2 Test voltage for
a. Cables including XLPE cables up to and including 33kV (WE.4.1)
b. Switchgear (WE.4.1)
c. Transformers up to 20kV (WE.4.3)

Exclusions: a. XLPE cables above 33kV (WE.4.1.3)
b. Transformers at 25kV and above (WE.4.3.1.a)
c. All VTs (WE.4.3.2)

Column 3 Alternative test voltage for substation apparatus

Column 4 Test voltage to be applied where insulation tests are required

Column 5 AC test voltage, either VLF or Power frequency, for EHV XLPE cables.

Columns 6&7 Test Voltage to be applied across the break of a vacuum interrupter (WE.4.2.2)

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(b) Testing to be carried out by the Supplier

The supplier shall propose to CE Electric UK: test arrangements and combinations, test values, test pass/fail criteria, tolerances applicable and associated test record content and layout for each of the tests listed below.

1. All protection and metering VTs and CTs shall be primary injection tested, at all available ratios, to prove ratio and polarity.
2. All trip coils shall be primary and secondary injected to test operation of the trip coil and subsequent operation of the circuit breaker.
3. Self powered relays shall have a factory default setting applied and a primary or secondary injection test carried out to determine and record the current pickup and 2x setting time values.
4. All secondary wiring shall be insulation resistance tested.
5. All fault flow indicator CTs shall be primary injected to prove ratio and polarity and the indicator unit shall be tested for detection and operation for both phase-phase and for phase-earth faults.
6. Power frequency tests shall be applied, using the criteria specified in CE Electric UK's Operational Practice Manual as included in part (a) of this Appendix.
Tests shall prove all combinations of operational condition: between each phase, between phase and earth and across all isolation gaps.
7. Tests shall be carried out, after all assembly operations have been completed, to confirm the phase relationship between primary conductors and all VDS, or VPIS, outputs and displays.
8. Partial discharge testing, preferably of the complete assembly. But at least of critical components, with the agreement with CE Electric UK.

With every switch panel the supplier shall provide CE Electric UK with: copies of the tests arrangements and test values applied, test results and associated test records for each of the above tests.

For all metering CTs and VTs; the supplier shall also provide CE Electric UK with copies of the CT and/or VT manufacturers' magnetising curve and ratio test certificates.

A complete set of printed, hard copies of these records shall be shipped in the associated LV control cabinet.

Electronic copies of these records shall also be supplied, if requested by CE Electric UK.

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Appendix 3 – Overall Summary Declaration of Compliance with NPS/003/014.

Switchgear range name and voltage range:			
NPS/003/014 Clause		Compliance (Yes or No)	Details/Remarks/Description <i>For all clauses; please write a short summary in this column</i>
3.1.1	ENA Assessed		ENA NOC ref:
3.1.2	Indoor or Outdoor design		
3.1.3	Level of remote control provision as standard		
3.2	Options for Time Limit Fuses and/or self powered relay protection		
3.2.1	Relays		
3.3	Metering (see Appendix1)		
3.4	Test voltages		
3.5	Physical interference		
3.6	Variations from ENATS 41-36:		
1.4.2.1	95kV BIL for 12kV eqpt		
1.4.4.1	Busbars >=630A		
1.4.4.1	Short-time withstand		
1.4.8	Remote Control; 24V DC supply		
1.4.8	Emergency (shunt) trip 110V AC and/or 30V DC		
1.5.1	Oil insulated items		
1.5.2	SF6 topping up		
1.5.8	HV metered supply units have remote (shunt trip)		
1.5.9	Gas pressure gauge as standard		
1.5.13.2	All indoor units are IPX1 or IPX2		
1.5.15	Gas leakage <=0.5% p.a.		
1.5.101	Internal Arc options		
1.5.102.103	Close coupling to 'cable connected' transformers		
1.5.103.1.101	Separable connectors		
1.5.201.3	VPIS/VDS on all circuits		
1.5.201.5	Ergonomic test devices		
1.10	UK Health & Safety legislation		
2.4	Line charging/breaking		
10.6.1	Integral CTs - standard		
10.6.1	Integral CT - options		
Appendix 2	Testing arrangements & values & record provision		

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Appendix 4 – Individual Unit Summary Technical Details.

One column to be completed for EACH unit/variant type listed in Appendix 1, including metering units.

	NOTE: This column contains example entries only		
Unit Type (for RMU; as default use 2xSwitch & 1xCircuit Breaker example)	Close Coupled RMU Non-extensible. Switch / CB / Switch	Unit 1	Unit 2
Manufacturers Type Reference	RMUABC-3a MkII		
Rated Voltage	12kV		
Rated Normal Current for switching devices (for RMUs: e.g. xA/yA/zA)	630 / 200 / 630 A		
Rated Short time Current	20 kA (1s)		
Impulse Withstand Voltage (BIL)	95kV		
Rated Normal Current for busbar	630A		
Indoor or Outdoor design	Outdoor		
Rating of Earthing Device(s)	Feeder E/Sw = 20kA CB E/Sw = 2kA		
Assigned number of fault-make operations, at full rating, before maintenance of each switching device (inc feeder earths)	Switch = 500 CB = 1000		
Assigned number of fault-break operations before maintenance of each switching device	Switch = 0 CB = 1000		
Physical Aspects			
Width of Unit	1100 mm		
Depth of Unit	1100 mm		
Height of Unit	1500 mm		
Weight (complete with filling medium)	400 kg		
Height of Centre of Gravity	850 mm		
Distance of Centre of Gravity from Rear of Unit/Tx Mounting Flange	450 mm		
Distance of Centre of Gravity from LHS of unit	550 mm		
Environmental			
Weight of SF ₆ , or oil, per unit	2.5 kg SF ₆		
Weight of SF ₆ , or oil, per chamber	Sw/CB/Sw common tank 2 kg SF ₆ Busbars 0.5 kg SF ₆		

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IEC Pressure (or density) significant values	NOTE: This column contains example entries only		
Rated Filling Pressure, or density, for Insulation and/or Switching P_{re} – IEC62271-1 3.6.5.1 (assembly) <i>Declaration Required for EACH gas zone/chamber</i>	Sw/CB/Sw common tank = 0.75 barG Busbars = 0.75 barG		
Rated Filling Pressure, or density, for Operation P_m – IEC62271-1 3.6.5.2 (control device) <i>Declaration Required for EACH gas zone/chamber</i>	As above, because all in common tank.		
Alarm Pressure, or density, for Operation P_{am} – IEC62271-1 3.6.5.4 - pressure at which a monitoring signal may be provided. P_{am} – IEC60694 3.6.4.4 - pressure at which a monitoring signal may be provided to indicate that replenishment may be necessary in a relatively short time. <i>Declaration Required for EACH gas zone/chamber</i>	Sw/CB/Sw common tank = 0.75 barG Busbars = 0.75 barG		
Minimum Functional Pressure, or density, for Operation P_{mm} – IEC62271-1 3.6.5.6 <i>Declaration Required for EACH gas zone/chamber</i>	Sw/CB/Sw common tank = 0.75 barG Busbars = 0.75 barG		
Alarm Pressure, or density, for Insulation and/or Switching P_{ae} – IEC62271-1 3.6.5.3 - pressure at which a monitoring signal may be provided. P_{ae} – IEC60694 3.6.4.3 - pressure at which a monitoring signal may be provided to indicate that replenishment may be necessary in a relatively short time. <i>Declaration Required for EACH gas zone/chamber</i>	Sw/CB/Sw common tank = 0.75 barG Busbars = 0.75 barG		
Minimum Functional Pressure, or density, for Operation and/or Switching P_{me} – IEC62271-1 3.6.5.5 - pressure at which and above which rated characteristics of switchgear and controlgear are maintained <i>Declaration Required for EACH gas zone/chamber and function.</i>	Sw/CB/Sw common tank CB = 0.6 barG Busbars = 0.6 barG		

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Practical Statement of Significant Pressures & Consequences	NOTE: This column contains example entries only		
Normal operating pressure (or density)	X barG		
First Alarm Level and details of any associated alarm contacts and details of implications for serviceability	Y barG Indicated on green/red zoned pressure gauge. When in gauge needle is in red area; switchgear cannot be guaranteed to provide rated characteristics. If has positive pressure then 1 set N/O contacts & 1 set N/C contacts.		
Second Alarm Level and details of any associated alarm contacts and details of implications for serviceability.	Not provided as standard; Option for ...		
Pressure (or density) at, or above which, rated characteristics for use as a Point of Isolation are maintained. Minimum Functional Pressure; I.e. disconnecter duty.	a bar G		
Pressure (or density) at or above which, rated characteristics for Breaking Fault Current are maintained. I.e. CB duty.	b bar G		
Pressure (or density) at or above which, rated characteristics for Making/Breaking Normal Load Current and Making Fault Current are maintained. I.e. Switch duty.	c bar G		
Pressure (or density) at or above which, rated characteristics for BIL are maintained.	d bar G		
Pressure (or density) at or above which, rated characteristics, except BIL, are maintained; allowing the units to remain in service, without carrying out switching duties. I.e. Busbar duty.	e bar G		
Under what operational conditions is it safe to top up the gas?			

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Cable Terminations			
Height of Cable Gland Plate above Floor Level	0 mm		
Cable Bushing Type	Outer cone, IECxxx, type Z		
Distance from Gland Plate to Cable Bushing centre/Lug Position	500 mm		
Cable Bushing - Phase/Phase distance between centres	250 mm		
Cable Bushing - Minimum Clearance Phase/Earth	N/A - solid, screened connectors		
Cable terminations and switchgear interface. <i>Please confirm the make and type of cable termination and cable combinations type tested with the switchgear</i>	(1) 185mm xyz single core cable from CableCo terminated with hjk separable connector kit from CableTermCo. (2) 185mm rst three core cable from CableCo terminated with rty cable termination kit from CableFixCo and uvw cable lugs from LugCo		

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Appendix 5a – ENATS 41-36 Technical Compliance Schedule.

ENATS 41-36 ANNEX D1 Common Clause SELF CERTIFICATION CONFORMANCE DECLARATION

PART 1 – COMMON CLAUSES

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 1, 'COMMON CLAUSES'

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 1 and the clauses of the aforementioned Standards relevant to common specifications for high-voltage switchgear and control gear standards. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Ratings:

Date:

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Confor- mance code	ENATS 41-36 - Part 1 Clause / Sub- clause	Requirement	Confor- mance code	Remarks
1	1	General		1.1	General		
2	2	Normal and special service conditions		1.2	Normal and special service conditions		
				1.2.1.1	Class minus 5 indoor		
				1.2.1.2	Class minus 25 outdoor		
				1.2.1.2	Class 10 – ice coating		
				1.2.1.2	Class III – pollution level		
				1.2.1.2	Influence of solar radiation		
3	3	Definitions		1.3	Definitions		
4	4	Ratings		1.4	Ratings		
4.1	4.1	Rated voltage		1.4.1	Rated voltage		
4.2	4.2	Rated insulation level		1.4.2	Rated insulation level		
				1.4.2.1	Disconnectors (0 bar gauge)		
				1.4.2.2.	Provision for cable tests		
4.3	4.3	Rated frequency		1.4.3	Rated frequency		
4.4	4.4	Rated normal current and temperature rise		1.4.4	Rated normal current and temperature rise		
				1.4.4.1	Rated normal current		
4.5	4.5	Rated short-time withstand		1.4.5	Rated short-time withstand		
4.6	4.6	Rated peak withstand current		1.4.6	Rated peak withstand current		
4.7	4.7	Rated duration of short circuit		1.4.7	Rated duration of short circuit		
4.8	4.8	Rated supply voltage of closing and opening devices and of auxiliary and control circuits		1.4.8	Rated supply voltage of closing and opening devices and of auxiliary and control circuits		
4.9	4.9	Rated supply frequency of closing and opening devices and of auxiliary circuits		1.4.9	Rated supply frequency of closing and opening devices and of auxiliary circuits		
4.10	4.10	Rated pressure of compressed gas supply for insulation and/or operation		1.4.10	Rated pressure of compressed gas supply for insulation and/or operation		
	4.10.1	Rated filling level (of fluid-filled compartments					
5.1	5.1	Requirements for liquids		1.5.1	Requirements for liquids		
				1.5.1	Oil level indication		
				1.5.1	Drain plugs, indicators, valves		
				1.5.1	Controlled gasket compression		
				1.5.1	No communicating bolts		
				1.5.1	BS 148		
				1.5.1	Bolt access		

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IEC 60694, IEC 62271-200			ENATS 41-36			Remarks
Clause / Sub-clause IEC60694 IEC62271-200	Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	
			1.5.1	Breather design/position (IP3XDW)		
5.2	5.2	Requirements for gasses	1.5.2	Requirements for gasses		
			1.5.2	Gas filling valve		
			1.5.2	Recycled SF6		
5.3	5.3	Earthing of switchgear and control gear	1.5.3	Earthing of switchgear and control gear		
			1.5.3	Earthing conductor		
			1.5.3	Earthing terminals.		
			1.5.3	Earthing conductor coupling		
			1.5.3	Withdrawable /removable parts earth connection.		
			1.5.3	Cable sheath earth connection		
			1.5.3	Relay/instrument case earthing		
			1.5.3	Specific means for earthing		
			1.5.3	Frame-earth busbar protection		
5.4	5.4	Auxiliary and control equipment	1.5.4	Auxiliary and control equipment		
			1.5.4.1.3	Degrees of protection – LV terminals		
			1.5.4.4.4.4	ENATS 50-19		
			1.5.4.4.4.4	Identification		
			1.5.4.4.5.1	Segregation (>125V).		
			1.5.4.4.5.1	Interchangeable - identical		
			1.5.4.4.5.1	Conductor material/size.		
			1.5.4.4.5.1	HV compartment segregation.		
			1.5.4.4.5.1	Actuator control/indication		
			1.5.4.4.5.1	Micro switches		
			1.5.4.4.5.2	Terminals/terminations reliability 50 breaks		
			1.5.4.4.5.2	CT terminal blocks – screw clamp with spring (ENATS 50-18 type B)		
5.5	5.5	Dependent power operation	1.5.5	Dependent power operation		
			1.5.5	Positively driven contacts		
			1.5.5	Movement gap withstand voltage		
			1.5.5	Maintenance / slow operation		
			1.5.5	Labelled - maintenance		
5.6	5.6	Stored energy operation	1.5.6	Stored energy operation		
			1.5.6	Sub-clause 1.5.5 applicable plus the following		
			1.5.6	Main contact movement		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	Remarks
				1.5.6	Dedicated handle		
				1.5.6	Handle direction indication		
				1.5.6	Handle release and stowed		
				1.5.6	Motor actuator fitting		
				1.5.6	Motor actuator disconnection		
				1.5.6	Actuator 'in step' (methods a or b)		
				1.5.6	Manual charging motor-charge		
				1.5.6	Max/min handle heights		
				1.5.6	Re-charge closing springs		
				1.5.6	Spring charge indication		
				1.5.7	Manual operation		
				1.5.7	Handles and padlocking accessible from front		
				1.5.7	Handle storage facilities		
5.7	5.7	Independent manual operation		1.5.7.1	Independent manual operation		
				1.5.7.1	Sub-clause 1.5.6 applicable plus the following		
				1.5.7.1	Inhibit closing spring charge in closed position		
				1.5.7.1	No stored energy from Incomplete operation		
				1.5.7.1	Anti-reflex =>3 secs (manual)		
				1.5.7.1	Anti-reflex =>3 secs (actuator)		
				1.5.7.2	Dependent manual operation. - as 1.5.7.1 plus: Inhibit op handle removal		
5.8	5.8	Operation of releases		1.5.8	Operation of releases		
				1.5.8	Local manual release		
				1.5.8	Operation outside switchroom		
				1.5.8	No movement of spring charge handle.		
5.9	5.9	Low and high-pressure interlocking and monitoring devices		1.5.9	Low and high-pressure interlocking and monitoring devices		
				1.5.9	Pressure/density gauge/indicator		
				1.5.9	20°C filling mark		
				1.5.9	Green/red, Go/No go		
				1.5.9	Single/two stage pressure switch		
				1.5.9	36kV equipment – a) & c)		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Confor- mance code	ENATS 41-36 - Part 1 Clause / Sub- clause	Requirement	Confor- mance code	Remarks
				1.5.9	Temperature fluctuations		
5.10	5.10	Nameplates		1.5.10.1	Nameplates		
				1.5.10.1	Internal arc test Fig		
				1.5.10.101	Labelling		
				1.5.10.101.1	Safety signs BS 5499		
				1.5.10.101.1	Durable/non-fading		
				1.5.10.101.1	Contrast with background		
				1.5.10.101.1	In accordance with Table 1.4		
				1.5.10.101.1	Symbols to Annex C		
				1.5.10.101.1	BS381C or RAL colours		
				1.5.10.101.2	Phase identification		
				1.5.10.101.3	Circuit labels to Fig 1		
				1.5.10.101.3	Additional labels to Fig 2		
				1.5.10.101.3	Repeat labels		
				1.5.10.101.3	Safely detachable		
				5.11	5.11	Interlocking devices	
				1.5.11	No removal of covers when part of interlock/padlock facility		
				1.5.11.101	Interlocking devices		
				1.5.11.101.1	General		
				1.5.11.101.1	Interlocking devices- Mechanical, key, electro-mechanical		
				1.5.11.101.2	Test access – interlocks a) to d)		
				1.5.11.102	Padlocking facilities		
				1.5.11.102	Size of padlock		
				1.5.11.102.1	Safety padlocks – facilities a) to c)		
				1.5.11.102.1	Single padlock for electrical and mechanical		
				1.5.11.102.1	Electrical/Electro-mechanical ‘FMECA’		
				1.5.11.102.1	Inhibit facia removal		
				1.5.11.102.1	Warning label		
				1.5.11.102.2	Operational padlocking		
1.5.11.102.2	Facilities (a) to (e)						
5.12	5.12	Position indication		1.5.12	Position indication		
				1.5.12	Positively driven mechanical		
				1.5.12	Output side of mechanism		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Confor- mance code	ENATS 41-36 - Part 1 Clause / Sub- clause	Requirement	Confor- mance code	Remarks
				1.5.12	Inscribed as Table 1.4		
				1.5.12	Mimic diagram symbols–Annex C		
				1.5.12	One indicator visible		
5.13	5.13	Degrees of protection by enclosures		1.5.13	Degrees of protection by enclosures		
				1.5.13.1	Hazardous parts / solid foreign objects IP4X,IP3X,IP3XD		
				1.5.13.1	Doors open IP2X		
				1.5.13.1	Requirements a)		
				1.5.13.2	Ingress of water		
				1.5.13.2	IP3XDW		
				1.5.13.2	Weather proofing test		
				1.5.13.2	IP34D - pole mounted		
				1.5.13.2	Material/water lodging		
				1.5.13.3	Mechanical impact 2J - indoor		
				1.5.13.3	Mechanical impact 5J - outdoor		
5.14	5.14	Creepage distances		1.5.14	Creepage distances and environmental considerations		
				1.5.14	Outdoor - class 3 –IEC 60815		
				1.5.14	Insulating system design		
				1.5.14	30 year life		
				1.5.14	Condensation/heaters		
				1.5.14	Shrouding in air filled cable box		
5.15	5.15	Gas and vacuum tightness		1.5.15	Gas and vacuum tightness		
				1.5.15	Leakage rate =< 1% per year (closed pressure) 30 year life expected		
				1.5.15	30 year life expected (sealed pressure)		
5.16	5.16	Liquid tightness		1.5.16	Liquid tightness		
5.17	5.17	Flammability		1.5.17	Flammability		
5.18	5.18	EMC		1.5.18	EMC		
	5.101	Internal fault		1.5.101	Internal fault		
				1.5.101	Class IAC		
				1.5.101	a) Metal enclosed – class IAC AF		
				1.5.101	b) Pole mounted – class IAC C		
				1.5.101	c) Air		
				1.5.101	d) 1 second duration		
				1.5.101	e) Prospective test current		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	Remarks
				1.5.101	f) Criteria of acceptance		
				1.5.101	g) Test arrangement		
				1.5.101	h) Compartments tested		
				1.5.101	Cable box prospective current		
	5.102	Enclosure		1.5.102	Enclosure		
				1.5.102.1	General		
				1.5.102.1	Support weight of personnel		
				1.5.102.1	Identify areas not safe to stand		
				1.5.102.1	Safe access - CDM Regs 1994		
				1.5.102.1	Lifting facilities		
				1.5.102.1	Integral step (150kg)		
				1.5.102.1	No communicating holes		
				1.5.102.2	Covers and doors		
				1.5.102.2	Interlock controlled accessible types for test access and provided with locking facilities		
				1.5.102.2	Controlled compression gaskets		
				1.5.102.2	No communicating holes		
				1.5.102.101	Surface preparation and coatings - ENATS 98-1		
				1.5.102.101	Colour		
				1.5.102.102	Foundation arrangements		
				1.5.102.102	Cable gland positions		
				1.5.102.102	Floor fixing (M12 bolts)		
				1.5.102.103	Transformer mounting		
				1.5.102.103	Access (fig 16 ENATS 35-1)		
				1.5.102.103	Transformer circuit flange as fig3		
				1.5.102.103	Dimensional limitations a) to g)		
				1.5.102.103	600mm max projection		
				1.5.102.103	Adjustable support		
				1.5.102.103	Load distribution		
				1.5.102.103	Assembly instructions		
				1.5.102.104	Heater		
				1.5.102.104	Easily accessible		
	5.103	Compartments		1.5.103	Compartments		
				1.5.103.1	Service continuity class LSC2 (except RME)		
				1.5.103.1.101	Cable compartments		

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IEC 60694, IEC 62271-200			ENATS 41-36					
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub- clause	Requirement	Conformance code	Remarks	
				1.5.103.1.101	Separate cable compartment			
				1.5.103.1.101	Cable compartment-ENATS 12-11			
				1.5.103.1.101	Compartment/termination design - manufacturers			
				1.5.103.1.101	Min of two propriety cable terminations systems			
				1.5.103.1.101	Accommodation / compatibility			
				1.5.103.1.101	Method statement			
				1.5.103.2	Fluid filled compartments			
				1.5.103.2.2	Solar radiation influences			
				1.5.103.2.3	Tightness			
				1.5.103.2.4	Pressure relief to be provided			
				1.5.103.2.4	No burn-through			
			1.5.103.2.4	Satisfactory performance in Outdoor environment				
				1.5.103.3	Partitions and shutters			
				1.5.103.3.1	Partitions metallic – class PM			
					Shutters metallic - class PM			
					Individually operated			
					Independently padlockable closed			
					Open/close automatically			
					Provision for retaining open			
					Re-engagement of removable part Restores automatic operation			
				Colour to table 1.4				
	5.104	Removable parts						
	5.105	Provisions for dielectric tests on cables						
					1.5.201.1.1	Testing via primary ccts a) to g)		
					1.5.201.1.2	Testing via secondary ccts a) to c)		
					1.5.201.2	Test facilities provided a) or b)		
					1.5.201.2	Test connections/main circuit- DC test (table 1.5)		
					1.5.201.2	Test connections/main circuit- 200A		
					1.5.201.2	Test terminals to cable =<500 microhms		
					1.5.201.2	Security of test contacts		
					1.5.201.2	Access to provide for safe working		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	Remarks
				1.5.201.2	Test point on mimic		
				1.5.201.3	Fixed equipment – VDS or VPIS		
				1.5.201.3	Withdrawable equipment - VDS		
				1.5.201.3	Min 60mm dia test access		
				1.5.201.4	Test access cover – 1.5.102.2)		
				1.5.201.4	“EARTH ON” for test access		
				1.5.201.4	Interlock on cable compartment		
				1.5.201.4	Test access at front		
				1.5.201.4	Physical indication of test access open (amber lamp for open, white lamp for closed)		
				1.5.201.4	Single lamp push to test facility		
				1.5.201.4	“EARTH ON” indication		
				1.5.201.4	No access to compartment containing live HV conductors.		
				1.5.201.4	Position to avoid water/debris ingress		
				1.5.201.4	Inhibit close of test access with test device inserted		
				1.5.201.4	Inhibit closing of disconnector or compromise POI with test access open		
				1.5.201.5	Test device security a) to c)		
				1.5.201.5	100 connections/disconnections		
				1.5.201.5	Test device identification		
				1.5.201.5	Test device container		
				1.5.202	Busbars - same current rating		
				1.5.202	Extension busbar trunking		
				1.5.202	Standard length		
				1.5.203	Conductor terminations		
				1.5.203	ENATS 41-16		
				1.5.203	BS 7354 clearances		
				1.5.203	BS 7354 clearances + 300mm		
6	6	Type tests		1.6	Type tests		
				1.6	Short circuit testing liaison (STL)		
				1.6	Criteria to pass lightning impulse		
				1.6	Dielectric (arrangement representative of cable termination systems in 1.5.103.1)		

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IEC 60694, IEC 62271-200				ENATS 41-36			
Clause / Sub-clause IEC60694 IEC62271-200		Requirement	Confor- mance code	ENATS 41-36 - Part 1 Clause / Sub- clause	Requirement	Confor- mance code	Remarks
				1.6	Most unfavourable arrangement		
				1.6	Production handle –most onerous in- service condition.		
				1.6	Partial discharge (ENATS 41-18 levels)		
				1.6	High temp or equivalent		
				1.6	Influence of solar radiation		
				1.6	Temperature rise at max solar gain		
7	7	Routine tests		1.7	Routine tests		
				1.7	Partial discharge (ENATS 41-18 Levels		
8	8	Guide to the selection of switchgear and controlgear		1.8	Guide to the selection of switchgear and controlgear		
9	9	Information to be given with enquiries, tenders and orders		1.9	Information to be given with enquiries, tenders and orders		
10	10	Rules for Transport, Storage, installation, operation and maintenance		1.10	Rules for Transport, Storage, installation, operation, maintenance and disposal		
				1.10	Safe methods for extending and/or replacement		
				1.10	Extension of corresponding types		
				1.10	Stable during storage/transport		
				1.10	Prevent water ingress		
				1.10	Temporary labels		
				1.10	Protect bushings		
				1.10	Maintenance design (BS 6626)		
				1.10	Handbook contents + storage		
11	11	Safety		1.11	Safety		

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Appendix 5b ENATS 41-36 ANNEX D2 Metal Enclosed Circuit-Breakers

SELF CERTIFICATION CONFORMANCE DECLARATION

PART 2 – METAL ENCLOSED CIRCUIT-BREAKERS

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 2, ' METAL ENCLOSED CIRCUIT-BREAKERS '

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 2 and the clauses of the aforementioned Standards relevant to metal enclosed circuit-breakers. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

This conformance declaration is to be completed in addition to the ENATS 41-36 – ANNEX D1– Self Certification Conformance Declaration Part 1, 'Common Clauses'.

For associated protection, instrumentation and metering equipment ENATS 41-36 – ANNEX D10 - Self Certification Conformance Declaration Part 10 is to be completed.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference :

Ratings:

Name:

Signature:

Date:

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IEC62271-100			ENATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub-clause	Requirement	Conform ance code	
1	General		2.1	General		
2	Service conditions		2.2	Service conditions		
3	Definitions		2.3	Definitions		
4	Ratings		2.4	Ratings		
4.101	Rated short-circuit breaking current		2.4.101	Rated short-circuit breaking current		
4.101.1	AC component of the rated short-circuit breaking current					
4.101.2	DC component of the rated short-circuit breaking current		2.4.101	Time constant (45ms)		
4.102	Transient recovery voltage related to the rated short-circuit breaking current		2.4.102	Transient recovery voltage related to the rated short-circuit breaking current		
4.102.1	Representation of TRV waves					
4.102.2	Representation of TRV					
4.102.3	Standard values of TRV related to the rated short-circuit breaking current		2.4.102	Table 1A of IEC 62271-100		
4.103	Rated short-circuit making current		2.4.103	Rated short-circuit making current		
4.104	Rated operating sequence		2.4.104	Rated operating sequence		
			2.4.104	O-0.3s-CO-15s-CO intended for auto-reclose		
			2.4.104	O-3min-CO-3min-CO not intended for auto-reclose		
4.105	Characteristics for short-line faults		2.4.105	Characteristics for short-line faults		
4.106	Rated out-of-phase making and breaking current		2.4.106	Rated out-of-phase making and breaking current		
4.107	Rated capacitive switching currents		2.4.107	Rated capacitive switching currents		
			2.4.107	Class C1		
4.107.1	Rated line-charging breaking		2.4.107	Line-charging breaking – Table 2.1 ENATS 41-36		
4.107.2	Rated cable-charging breaking		2.4.107	Cable-charging breaking - Table 2.1 ENATS 41-36		
4.108	Rated small inductive breaking current		2.4.108	Rated small inductive breaking current		
4.109	Rated time quantities		2.4.109	Rated time quantities		
4.109.1	Rated break time			Rated break time		
4.110	Number of mechanical operations		2.4.110	Number of mechanical operations		
			2.4.110	Class M2 (10000 Ops) (5000 Ops)		
			2.4.110	Class M1 (2000 Ops)		
4.111	Classification of circuit-breakers as a function of electrical endurance		2.4.111	Classification of circuit-breakers as a function of electrical endurance		

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IEC62271-100			ENATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub-clause	Requirement	Conform ance code	
5.1	Requirements for liquids in circuit-breakers		2.4.111	Class E2		
			2.4.111	Auto-reclosing		
			2.4.111	Non auto-reclosing		
			2.5.1	Requirements for liquids in circuit-breakers		
5.2	Requirements for gasses		2.5.1	Correct tank mounting.		
			2.5.1	Inhibit service/earth position without tank.		
			2.5.1	Correct contact/arc device positions		
5.3	Requirements for gasses		2.5.2	Requirements for gasses		
5.3	Earthing of circuit-breakers		2.5.3	Earthing of circuit-breakers		
			2.5.3.101	Earthing facilities		
			2.5.3.101.1	Short circuit making ability		
			2.5.3.101.1	CB or class E2 earthing switch		
			2.5.2.101.1	Kinematic chain test on CB used for earthing		
			2.5.3.101.1	CT short circuit facility		
			2.5.3.101.1	Earthing device ratings		
			2.5.3.101.2	Disconnecter / earthing switch		
			2.5.3.101.2	AC + DC withstand test		
			2.5.3.101.2	Class E0		
			2.5.3.101.2	Class M0		
			2.5.3.101.2	Number of mech ops		
			2.5.3.101.2	Selector arrangements (a to d)		
5.4	Auxiliary equipment		2.5.4	Auxiliary equipment		
			2.5.4	Isolating features		
			2.5.4	Secondary connections		
			2.5.4	Secondary circuit coupling for maintenance		
			2.5.4	Contact location/screening		
5.5	Dependent power closing		2.5.5	Dependent power closing		
5.6	Stored energy closing		2.5.6	Stored energy closing		
5.7	Independent manual		2.5.7	Independent manual		
5.8	Operation of releases		2.5.8.	Operation of releases		
			2.5.8.	Inhibit continuous open/close		
			2.5.8.	Padlockable device - RED		
			2.5.8.	Closing facility - BLACK		
			2.5.8.	Push buttons RED/BLACK		

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IEC62271-100			ENATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub-clause	Requirement	Conform ance code	
			2.5.8.	Remote operation socket		
			2.5.8.	Local/remote selector		
5.9	Low and high-pressure interlocking and monitoring devices		2.5.9	Low and high-pressure interlocking and monitoring devices		
5.10	Nameplates		2.5.10.1	Nameplates		
			2.5.10.1	Supplementary (a to c)		
			2.5.10.101	Labelling		
			2.5.10.101	Mimic diagram		
			2.5.10.101	Test points		
			2.5.10.101	VT /VT disconnection		
			2.5.10.101	Mimic colour/symbols (Annex C)		
			2.5.10.101	Pictogram for busbar earthing (Annex C)		
5.11	Interlocking devices		2.5.11	Interlocking / padlocking		
			2.5.11.101	Interlocking devices		
			2.5.11.101.1	Indoor circuit-breakers		
			2.5.11.101.1	Inhibit close of disconnectors with test access open		
			2.5.11.101.1	No CB trip on attempted isolation / disconnectors selection		
			2.5.11.101.1	No main circuit access		
			2.5.11.101.1	Safety shutters remain closed		
			2.5.11.101.1	Earthing device located		
			2.5.11.101.1	Inhibit disconnector closure		
			2.5.11.101.1	Move before BB earthing device		
			2.5.11.101.1	Inhibit circuit breaker trip		
			2.5.11.101.2	Outdoor circuit-breakers		
			2.5.11.101.2	Key interlock		
5.12	Position indication		2.5.12	Position indication		
5.13	Degrees of protection		2.5.13	Degrees of protection		
5.14	Creepage distances		2.5.14	Creepage distances		
5.15	Gas and vacuum tightness		2.5.15	Gas and vacuum tightness		
5.16	Liquid tightness		2.5.16	Liquid tightness		
5.17	Flammability		2.5.17	Flammability		
5.18	Electromagnetic compatibility		2.5.18	Electromagnetic compatibility		
5.101	Requirements for simultaneity of poles during single closing and single opening operations.					
5.102	General requirement for operation					
5.103	Pressure limits of fluids for operation					

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IEC62271-100			ENATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub-clause	Requirement	Conform ance code	
5.104	Vent outlets					
			2.5.201	Test facilities		
			2.5.202	Open-terminal circuit-breaker bushings		
			2.5.202	Prismatic gauge		
			2.5.202	Terminals (ENATS 41-16) BS 7354 design		
			2.5.203	Outdoor open-terminal circuit-breaker control facilities		
			2.5.203	Operation facilities		
			2.5.203	Selector switch		
			2.5.203	Close/open control switch		
			2.5.204	Open-terminal circuit-breaker mechanism cabinet		
			2.5.204	ON/OFF indicator		
			2.5.204	Inside and outside labels		
			2.5.204	Multicore terminal blocks		
			2.5.204	Power socket +RCD		
			2.5.205	Transformer mounting		
6	Type tests		2.6	Type Tests - Table 2.3		
6.1	General					
6.2	Dielectric tests					
6.3	Radio interference voltage tests					
6.4	Measurement of resistance of the main circuit					
6.5	Temperature rise tests					
6.6	Short-time withstand current and peak withstand current tests					
6.7	Verification of the degree of protection					
6.8	Tightness test					
6.9	Electromagnetic compatibility (EMC) tests					
6.101	Mechanical and environmental tests					
6.102	Miscellaneous provisions for making and breaking tests					
6.103	Test circuits for short-circuit making and breaking tests					
6.104	Short-circuit test quantities					
6.105	Short-circuit test procedure					

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IEC62271-100			ENATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub-clause	Requirement	Conform ance code	
6.106	Basic short-circuit test duties					
6.107	Critical current tests					
6.108	Single phase and double earth fault tests					
6.109	Short-line fault tests					
6.110	Out of phase making and breaking tests					
6.111	Capacitive current switching tests					
6.112	Special requirements for making and breaking tests on class E2 circuit-breakers.					
7	Routine Tests		2.7	Routine Tests		
7.1	Dielectric test on the main circuit					
7.2	Dielectric test on auxiliary and control circuits					
7.3	Measurement of resistance of the main circuit					
7.4	Tightness test					
7.5	Design and visual checks					
8	Guide to the Selection of Circuit-breakers for service		2.8	Guide to the Selection of Circuit-breakers for service		
9	Information to be Given with Enquiries, Tenders and Orders		2.9	Information to be Given with Enquiries, Tenders and Orders- schedule 2.1		
10	Rules for Transport, Storage, Erection and Maintenance		2.10	Rules for Transport, Storage, Erection and Maintenance		
			2.10	Contact erosion		
			2.10	Operation counter		
11	Safety		2.11	Safety		

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Appendix 5C ENATS 41-36 ANNEX D2 Metal Enclosed Circuit-Breakers

SELF CERTIFICATION CONFORMANCE DECLARATION

PART 2 – METAL ENCLOSED CIRCUIT-BREAKERS -TYPE TEST CONFORMANCE DECLARATION D2

Type tests for feeder or bus-section circuit-breaker including enclosure, disconnecter, VT, CTs and earthing switch as appropriate.

Manufacturer:

Product reference:

Name:

Signature:

Ratings

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
 - ENA/SAP to complete columns 1 to 4
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the table

Type test reports table based on ENATS 41-36 Table 2.3

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
Test Requirement	Specification & Standards	Rated value	Test req'd Y or N	Conformance*	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
1. Dielectric. Partial discharge $\leq 10\text{pC}$ For cable connected circuit-breakers, tests to be representative of two cable termination systems, in addition to switchgear manufacturer's own system if any.	IEC60694. Sub-clause 6.2, IEC62271-100. Sub-clause 6.2, IEC62271-200. Sub-clause 6.2. Tables 1.1a and 1.1b of ENATS 41-36. IEC 62271-200. Sub-clause 6.2.9 and annex BB, ENATS 41-18 Sub-clauses 1.5.103.1 and 1.6 of ENATS 41- 36.								
2. " - Busbars	"								
3. Voltage Withstand - Isolating Gap (provision for dielectric tests on cables).	IEC62271-200. Sub-clause 5.105. and 6.2.101 Sub-clause 1.4.2.2 of ENATS 41-36.								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification & Standards	Rated value	Test req'd Y or N	Conformance*	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
4.	DC Withstand Test on Test Devices, including all parts of main circuit which cannot be disconnected from the test connections.	IEC62271-200. Sub-clause 5.105. Sub-clause 1.5.201.1 of ENATS 41-36.								
5.	Insulation level - electrically stressed gap due to possible movement of earthing switch contacts	Sub-clause 1.5.6 of ENATS 41-36								
6.	Measurement of the resistance of main circuit – Panel	IEC 60694. Sub-clause 6.4, IEC62271-100. Sub-clause 6.4, IEC62271-200. Sub-clause 6.4.								
7.	“ - Busbars	“								
8.	Temperature Rise - Panel	IEC 60694. Sub-clause 6.5, IEC62271-100. Sub-clause 6.5, IEC62271-200. Sub-clause 6.5								
9.	“ - Busbars	“								
10	Short-time withstand current and peak withstand current tests - Circuit-breaker and Enclosure.(3sec short time)	IEC60694. Sub-clause 6.6, IEC62271-100. Sub-clause 6.6, IEC62271-200. Sub-clause 6.6.								
11	“ - Busbars	“								
12.	“ - Earthing Switch	IEC60694. Sub-clause 6.6, IEC62271-102. Sub-clause 6.6, IEC62271-200. Sub-clause 6.5.								
13.	“ - Single phase test of earth circuit (sub-clause 1.4.5 of ENATS 41-36)	IEC60694. Sub-clause 6.6, IEC62271-100. Sub-clause 6.6, IEC62271-200. Sub-clause 6.6.								
14.	Verification of protection. (Indoor – IP3X min) Weatherproofing for outdoor equipment (IP3XDW min) Mechanical impact (indoor – 2J, outdoor – 5J)	IEC60694. Sub-clause 6.7, IEC62271-100. Sub-clause 6.7, IEC62271-200. Sub-clause 6.7. IEC 529 Sub-clause 1.5.13 of ENATS 41-36.								
15.	Tightness test	IEC60694. Sub-clause 6.8, IEC62271-100. Sub-clause 6.8, IEC62271-200. Sub-clause 6.8.								
16.	EMC tests	IEC60694. Sub-clause 6.9,								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification & Standards	Rated value	Test req'd Y or N	Conformance*	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
17.	Mechanical operations - Circuit-breaker. a) Auto-reclosing circuit-breaker -min 5000 ops, preferably class M2 - 10,000 operating cycles, auto-reclosing sequences - Table 8, IEC 62271-100) b) Non auto-reclosing circuit-breaker - class M1 - 2000 operating cycles If used as earthing device - mechanical strength of kinematic chain between movable contacts and the position indicating device	IEC62271-100. Sub-clause 6.9, IEC62271-100. Sub-clause 6.101.2. IEC62271-200. Sub-clause 6.102 Sub-clause 2.4.110 of ENATS 41-36 IEC62271-102. Sub-clause 6.105 and Annex A								
18.	Mechanical operations - Earthing switch and disconnecter - Manual - 1000 operating cycles Disconnectors operating in conjunction with circuit - breaker –2000, (5000) or 10,000 operating cycles depending on class of circuit-breaker Mechanical strength of kinematic chain between movable contacts and the position indicating device.	IEC62271-102. Sub-clause 6.102 IEC62271-200. Sub-clause 6.102 IEC62271-102. Sub-clause 6.105 and Annex A								
19.	Mechanical operations-switching devices and removable parts-50ops - interlocks (mechanical and electro-mechanical - 50 ops	IEC62271-200. Sub-clause 6.102								
20.	Low temperature tests	IEC 62271-100. Sub-clause 6.101.3								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification & Standards	Rated value	Test req'd Y or N	Conformance*	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
21.	High temperature tests - subject to design (see clause 1.6)	IEC 62271-100. Sub-clause 6.101.3. Clause 1.6 of ENATS 41-36								
22.	Short-circuit making and breaking tests - Circuit-breaker class E2 Auto-reclosing circuit-breaker - tested for duty as specified in IEC62271-100. table 21, list 1. Non auto-reclosing circuit-breaker - tested in accordance with sub-clauses 6.112.1 and 6.106 of IEC 62271-100	IEC62271-100. Sub-clauses 6.102 to 6.106, and 6.112 IEC62271-200. Sub-clause 6.101 Sub-clause 2.4.111 of ENATS 41-36								
23.	Short-circuit making tests - Earthing switch (class E2 - Test duty 5 of IEC 60265 sub-clause 6.101.10 - 5 making ops).	IEC62271-102. Sub-clause 6.101, IEC60265: PART 1. Sub-clause 6.101.10								
24	Out of phase making and breaking current	IEC62271-100. Sub-clause 6.110, Sub-clause 2.4.106 of ENATS 41-36								
25.	Line-charging breaking current tests	IEC62271-100. Sub-clause 6.111 Sub-clause 2.4.107 of ENATS 41-36								
26.	Cable-charging breaking current tests	IEC62271-100. Sub-clause 6.111 Sub-clause 2.4.107 of ENATS 41-36								
27.	Internal Arc -- C B Chamber, C T Chamber, B B Chamber, Cable box. (36kV cable boxes using separable connectors – min 5kA).	IEC62271-200. Sub-clause 6.106. and Annex A Sub-clause 1.5.101 of ENATS 41-36								
28.	Gas-filled Compartment – Pressure Withstand	IEC62271-200. Sub-clause 6.103								
29.	Voltage presence indicating system (VPIS)	IEC 61958. Clause 6								
30.	Finish	Performance to ENATS 98-1								
31.	Process Control	ISO 9001								

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* Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests

Ct2 = Not fully independent witnessed tests

Ct3 = Self verified tests

Ct4 = Alternative tests / evidence offered

Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.

Ct6 = Not tested

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Appendix 5D ENATS 41-36 ANNEX D3 Metal Enclosed Switches SELF CERTIFICATION CONFORMANCE DECLARATION

PART 3 – METAL ENCLOSED SWITCHES

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 3, ' METAL ENCLOSED SWITCHES '

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 3 and the clauses of the aforementioned Standards relevant to metal enclosed switches. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

This conformance declaration is to be completed in addition to the ENATS 41-36 – ANNEX D1– Self Certification Conformance Declaration Part 1, 'Common Clauses'.

For associated protection, instrumentation and metering equipment ENATS 41-36 – ANNEX D10 - Self Certification Conformance Declaration Part 10 is to be completed.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Ratings:

Date:

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IEC60265-1, IEC 62271-102			ENATS 41-36 - Part 3			Remarks
Clause / Sub-clause IEC 62271 - 102	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
1	1	General	3.1	General		
			3.1	Switch disconnecter		
			3.1	Class E3 - IEC 600265		
			3.1	10 closing ops at rated s/c		
			3.1	≤ 24kV- Class M2 - 5000 ops		
			3.1	36Kv - Class M1 -1000 ops		
2	2	Normal service and special service conditions	3.2	Normal service and special service conditions		
3	3	Definitions	3.3	Definitions		
4	4	Ratings	3.4	Ratings		
	4.101	Rated short-circuit making current				
4.101		Rated mainly active load breaking current	3.4.101	Rated mainly active load breaking current		
4.102		Rated closed-loop breaking current	3.4.102	Rated closed-loop breaking current		
4.103		Rated no-load transformer breaking current	3.4.103	Rated no-load transformer breaking current		
4.104		Rated cable-charge breaking current	3.4.104	Rated cable-charge breaking current		
4.105		Rated line-charging breaking current	3.4.105	Rated line-charging breaking current		
4.112		Rated short-circuit making current	3.4.112	Rated short-circuit making current		
4.113		Rated breaking and making currents Class E3 / 10 x TD5	3.4.113	Rated breaking and making currents Class E3 / 10 x TD5		
			3.4.201	Rated Mechanical Endurance		
5	5	Design and construction	3.5	Design and construction		
5.1	5.1	Requirements for liquids	3.5.1	Requirements for liquids		
5.2	5.2	Requirements for gasses	3.5.2	Requirements for gasses		
5.3	5.3	Earthing of high voltage switches	3.5.3	Earthing of high voltage switches and earthing facilities		
			3.5.3	Class E2 earthing switch		
			3.5.3	Earthing switch ratings		

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IEC60265-1, IEC 62271-102			ENATS 41-36 - Part 3			Remarks
Clause / Sub-clause IEC 62271 - 102	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
5.4	5.4	Auxiliary and control equipment	3.5.4	Auxiliary and control equipment		
5.5	5.5	Dependent power closing	3.5.5	Dependent power closing		
5.6	5.6	Stored energy operation	3.5.6	Stored energy operation		
			3.5.6	Switch mechanisms		
			3.5.6	Earthing switch mechanism		
5.7	5.7	Independent manual operation	3.5.7	Independent manual operation		
5.8	5.8	Operation of releases	3.5.8	Operation of releases		
5.9	5.9		3.5.9	Low and high-pressure interlocking and monitoring devices		
5.10	5.10	Nameplates	3.5.10	Nameplates and labelling		
			3.5.10	Mimic diagram		
			3.5.10	Test points		
			3.5.10	Mimic symbols/colours - Annex C		
			3.5.10	Labels - typically Fig 6		
5.11	5.11	Interlocking devices	3.5.11	Interlocking devices and padlocking facilities		
			3.5.11	Move before earthing device		
			3.5.11	Interlocks (a) to (f)		
5.12	5.12	Position indicating	3.5.12	Position indicating		
5.13	5.13	Degrees of protection by enclosures	3.5.13	Degrees of protection by enclosures		
5.14	5.14	Creepage distance	3.5.14	Creepage distance		
5.15	5.15	Gas and vacuum tightness	3.5.15	Gas and vacuum tightness		
5.16	5.16	Liquid tightness	3.5.16	Liquid tightness		
5.17	5.17	Flammability	3.5.17	Flammability		
5.18	5.18	Electromagnetic compatibility	3.5.18	Electromagnetic compatibility		
5.101		Making and breaking operations				
	5.101	Special requirements for earthing switches				
5.102	5.102	Requirements for switch-disconnectors				
5.103	5.103	Mechanical strength				
5.104		Securing the position				

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IEC60265-1, IEC 62271-102			ENATS 41-36 - Part 3			Remarks
Clause / Sub-clause IEC 62271 - 102	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
	5.104	Operation of disconnectors and earthing switches – Position of the movable contact system and its indicating and signalling devices				
	5.104.1	Securing of position				
	5.104.2	Additional requirements for power-operated mechanisms				
	5.104.3	Indication and signalling of position				
	5.104.3.1	Indication of position				
	5.104.3.2	Electrical position signalling by auxiliary contacts				
5.105		Auxiliary contacts for signalling				
	5.105	Maximum force required for manual operation				
	5.105.1	Operation requiring more than one revolution				
	5.105.2	Operation requiring up to one revolution				
	5.106	Dimensional tolerances				
			3.5.201	Test facilities		
			3.5.201	Separate		
			3.5.201	integral		
			3.5.201	Test symbol indication		
			3.5.201	Test symbol - Annex C		
			3.5.202	Transformer mounting		
6		Type tests	3.6	Type Tests		
			3.6	Table 3.2		
6.1		General				
6.2		Dielectric tests				
6.3		Radio interference voltage tests				
6.4		Measurement of resistance of the main circuit				

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IEC60265-1, IEC 62271-102			ENATS 41-36 - Part 3			Remarks
Clause / Sub-clause IEC 62271 - 102	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
6.5	Temperature rise tests					
6.6	Short-time withstand current and peak withstand current tests					
6.7	Verification of the degree of protection					
6.8	Tightness test					
6.9	Electromagnetic compatibility (EMC) tests					
6.101	Making and breaking tests					
6.102	Mechanical operations test					
6.103	Operation under severe ice conditions					
7	Routine Tests		3.7	Routine Tests		
8	Guide to the Selection of high- voltage switches for service		3.8	Guide to the Selection of high-voltage switches for service		
9	Information to be Given with enquiries, tenders and orders		3.9	Information to be Given with enquiries, tenders and orders- schedule 3.1		
10	Rules for transport, storage, erection, operation and maintenance		3.10	Rules for transport, storage, erection, operation and maintenance		
11	Safety		3.11	Safety		

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Appendix 5E ENATS 41-36 ANNEX D3 Metal Enclosed Switches SELF CERTIFICATION CONFORMANCE DECLARATION

PART 3 – METAL ENCLOSED SWITCHES - TYPE TEST CONFORMANCE DECLARATION D3

Type tests for switch-disconnector including enclosure and earthing switch.

Manufacturer:

Product reference:

Name:

Signature:

Ratings

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
 - ENA/SAP to complete columns 1 to 4
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the table

Type test reports table based on ENATS 41-36 Table 3.2

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conform ance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
1	Dielectric switch- - Panel, including disconnecter and earthing switch as appropriate. Partial discharge ≤ 10pC For cable connected switches, tests to be representative of two cable termination systems, in addition to switchgear manufacturer's own system if any	IEC 60694. Sub-clause 6.2, IEC 60265: PART 1 . Sub-clause 6.2, IEC 62271-200. Sub-clause 6.2. Tables 1.1a and 1.1b of ENATS 41-36. IEC 62271-200. Sub-clause 6.2.9 and Annex FF ENATS 41-18 Sub-clauses 1.5.103.1 and 1.6 of ENATS 41-36								
2	" - Busbars	"								
3	Voltage Withstand - Isolating Gap (provision for dielectric tests on cables).	IEC 62271-200. Sub-clause 5.105 and 6.2.101. Sub-clause 1.4.2.2 of ENATS 41-36.								
4	DC Withstand Test on Test Devices,	IEC 62271-200. Sub-clause 5.105.								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conform ance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
	including all parts of main circuit which cannot be disconnected from the test connections	Sub-clause 1.5.201.1 of ENATS 41-36.								
5	Insulation level - electrically stressed gap due to possible movement of earthing switch contacts	Sub-clause 1.5.6 of ENATS 41-36								
6	Measurement of the resistance of main circuit – Panel	IEC 60694. Sub-clause 6.4, IEC 60265: PART 1 . Sub-clause 6.4, IEC 62271-200. Sub-clause 6.4								
7	“ - Busbars	“								
8	Temperature Rise - Panel	IEC 60694. Sub-clause 6.5, IEC 60265: PART 1 . Sub-clause 6.5, IEC 62271-200. Sub-clause 6.5								
9	“ - Busbars	“								
10	Short-time withstand current and peak withstand current tests - switch-disconnector and Enclosure.(3sec short time)	IEC 60694.Sub-clause 6.6, IEC 60265: PART 1 . Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
11	“ - Busbars	“								
12	“ - Earthing Switch	IEC 60694.Sub-clause 6.6, IEC 62271-102. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
13	“ -Single phase test of earth circuit	IEC 60694.Sub-clause 6.6, IEC 62271-102. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
14	Verification of protection. (indoor – IP3X min) Weatherproofing for outdoor equipment. (IP3XDW min) Mechanical impact (indoor 2J, outdoor 5J)	IEC 60694.Sub-clause 6.7, IEC 60265: PART 1 . Sub-clause 6.7, IEC 62271-200. Sub-clause 6.7 IEC 529 Sub-clause 1.5.13 of ENATS 41-36.								
15	Tightness test	IEC 60694.Sub-clause 6.8, IEC 60265: PART 1 . Sub-clause 6.8, IEC 62271-200. Sub-clause 6.8.								
16	EMC tests	IEC 60694.Sub-clause 6.9, IEC 60265: PART 1 . Sub-clause 6.9,								
17	Short-circuit making and breaking tests - Switch-disconnector class E3 (7.2kV, 12kV, 24kV Test duty 5 = 10ops) (36kV test duty 5 = 5 ops)	IEC 60265: PART 1 . Sub-clauses 6.101, (TD 1 to 5, Table 5) IEC 62271-200. Sub-clause 6.101 Sub-clause 3.4.113 of ENATS 41-36								
18	Short-circuit making tests - Earthing switch (class E2) Test duty 5 of IEC 60265 sub-clause	IEC 62271-102. Sub-clause 6.101, IEC60265. Sub-clause 6.101.10								

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	1. Test Requirement	2. Specification and standards	3. Rated value	4. Test req'd Y or N	5. Conform ance *	6. Test value	7. Date of test	8. Test station Report / Cert No	9. Witness I, M or ENA **	10. Remarks
	6.101.10 - 5 making operations.									
19	Mechanical operations - Switch-disconnector - class M2 General purpose switch - 5000 operations. (including mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC 60265: PART 1. Sub-clause 6.102.4, IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
20	Mechanical operations - Earthing switch (including mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC 62271-102. Sub-clause 6.102 IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
21	Mechanical operations -Switching devices and removable parts- 50 ops -Interlocks-50 ops(mechanical and electro-mechanical)	IEC 62271-200. Sub-clause 6.102								
22	Low temperature tests	IEC 62271-102. Sub-clause 6.104								
23	High temperature tests- subject to design (see clause 1.6)	IEC 62271-102. Sub-clause 6.104, Clause 1.6 of ENATS 41-36								
24	Cable-charging breaking current tests	IEC 60265: PART 1. Sub-clause 6.101.8.4(f)								
25	Line-charging breaking current tests	IEC 60265: PART 1. Sub-clause 6.101.8.4(g)								
26	Internal Arc -- Switch-disconnector chamber, BB Chamber, Cable box (36kV cable boxes using screened separable connectors – min 5kA.)	IEC 62271-200. Sub-clause 6.106. Sub-clause 1.5.101 of ENATS 41-36								
27	Gas-filled Compartment Pressure Withstand	IEC 62271-200. Sub-clause 6.103								
28	Voltage presence indicating system (VPIS)	IEC 61958. Clause 6								
29	Voltage detecting device (VDS)	IEC 61243-5. Clause 5								
30	Finish	Performance to ENATS 98-1								
31	Process Control	ISO 9001								

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* Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The test conforms fully with the requirements of this clause
Cs2 = The test conforms partially with the requirements of this clause
Cs3 = The test does not conform to the requirements of this clause
Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests
Ct2 = Not fully independent witnessed tests
Ct3 = Self verified tests
Ct4 = Alternative tests / evidence offered
Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.
Ct6 = Not tested

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Appendix 5F ANNEX D4 Metal Enclosed Switch-Fuse Combinations SELF CERTIFICATION CONFORMANCE DECLARATION

PART 4 – METAL ENCLOSED SWITCH-FUSE COMBINATIONS

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 4, ' METAL ENCLOSED SWITCH-FUSE COMBINATIONS '

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 4 and the clauses of the aforementioned Standards relevant to metal enclosed switch-fuse combinations. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

This conformance declaration is to be completed in addition to the ENATS 41-36 – ANNEX D1– Self Certification Conformance Declaration Part 1, 'Common Clauses'.

For associated protection, instrumentation and metering equipment ENATS 41-36 – ANNEX D10 - Self Certification Conformance Declaration Part 10 is to be completed.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Ratings:

Date:

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IEC62271-105			ENATS 41-36 - Part 4			Remarks
Clause / Sub-clause	Requirement	Conformance code	Clause / Sub-clause	Requirement	Conformance code	
1	General		4.1	General		
			4.1	Switch-fuse / Fuse switch		
2	Normal service and special service conditions		4.2	Normal service and special service conditions		
3	Definitions		4.3	Definitions		
4	Ratings		4.4	Ratings		
4.101	Rated short-circuit breaking current		4.4.101	Rated short-circuit breaking current		
4.102	Rated transient recovery voltage		4.4.102	Rated transient recovery voltage		
4.103	Rated short-circuit making current		4.4.103	Rated short-circuit making current		
4.104	Rated transfer current (striker operation)		4.4.104	Rated transfer current (striker operation)		
4.105	Rated take-over current for release-operated combinations		4.4.105	Rated take-over current for release-operated combinations		
5	Design and construction		4.5	Design and construction		
5.1	Requirements for liquids		4.5.1	Requirements for liquids		
5.2	Requirements for gasses		4.5.2	Requirements for gasses		
5.3	Earthing of switch-fuse combinations		4.5.3	Earthing of switch-fuse combinations and earthing facilities		
				1 or 2 earthing switches		
				a) Incoming circuit earthing		
				b) No incoming circuit earthing		
				c) Outgoing circuit earthing		
				d) No outgoing circuit earthing		
				Class E2 earthing switches		
				Earthing switch ratings		
				No rating impairment		
5.4	Auxiliary and control equipment		4.5.4	Auxiliary and control equipment		
5.5	Dependent power closing		4.5.5	Dependent power closing		
5.6	Stored energy closing		4.5.6	Stored energy closing		
			4.5.6	Switch-fuse - independent manual		
			4.5.6	Earthing switch-independent manual		
5.7	Independent manual operation		4.5.7	Independent manual operation		
5.8	Operation of releases		4.5.8	Operation of releases		
				Fuse operation opens 3 phases		
				Manual trip		
				Trip - label		
5.9	Low and high-pressure interlocking and monitoring devices		4.5.9	Low and high-pressure interlocking and monitoring devices		

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IEC62271-105			ENATS 41-36 - Part 4			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
5.10	Nameplates		4.5.10	Nameplates and labelling		
			4.5.10	Labels - typically Fig 7		
			4.5.10	Test points		
			4.5.10	Mimic symbols/colours - Annex C		
			4.5.10	Mimic diagram		
5.11	Interlocking devices		4.5.11	Interlocking devices and padlocking facilities		
			4.5.11	Move before earthing device		
				Incoming / outgoing earthing switch interlocks (independent)		
			4.5.11	Interlocks (a) to (i)		
5.12	Position indicating		4.5.12	Position indicating		
5.13	Degrees of protection by enclosures		4.5.13	Degrees of protection by enclosures		
				Live parts shielded		
				Prevent tools falling (live parts)		
				Captive, hexagonal screws/nuts		
5.14	Creepage distance		4.5.14	Creepage distance		
5.15	Gas and vacuum tightness		4.5.15	Gas and vacuum tightness		
5.16	Liquid tightness		4.5.16	Liquid tightness		
5.17	Flammability		4.5.17	Flammability		
5.18	Electromagnetic compatibility		4.5.18	Flammability		
5.101	Linkages between the fuse striker(s) and the switch release		4.5.101	Linkages between the fuse striker(s) and the switch release		
5.102	Low over-current conditions (long fuse pre-arcing time conditions)		4.5.102	Low over-current conditions (long fuse pre-arcing time conditions)		
			4.5.201	Test facilities		
			4.5.201	With class E2 earthing switches		
			4.5.201	Separate		
			4.5.201	integral		
			4.5.201	Test symbol indication		
			4.5.201	Test symbol - Annex C		
			4.5.202	Transformer mounting		
			4.5.203	Fuse-link arrangement		
			4.5.203	Safe removal and replacement		
			4.5.203	Access step		
			4.5.203	Isolation		
			4.5.203	Visible isolation / earthing		
			4.5.203	Oil splash protection		
6	Type tests		4.6	Type Tests		

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IEC62271-105			ENATS 41-36 - Part 4			Remarks
Clause / Sub-clause	Requirement	Conform ance code	Clause / Sub- clause	Requirement	Conform ance code	
			4.6	Table 4.1		
6.1	General					
6.2	Dielectric tests					
6.3	Radio interference voltage tests					
6.4	Measurement of resistance of the main circuit					
6.5	Temperature rise tests					
6.6	Short-time withstand current and peak withstand current tests					
6.7	Verification of the degree of protection					
6.8	Tightness test					
6.9	Electromagnetic compatibility (EMC) tests					
6.101	Making and breaking tests					
6.102	Mechanical operations test					
6.103	Mechanical shock test on fuses					
6.104	Thermal test with long pre-arcing time of fuse					
7	Routine Tests		4.7	Routine Tests		
7.101	Mechanical operating tests					
8	Guide to the Selection of high-voltage switch-fuse combinations for transformer protection		4.8	Guide to the Selection of high-voltage switch-fuse combinations for transformer protection		
			4.8.1	Fuse-links - BS 2692, IEC 60282		
			4.8.1	Fuse-links - BS 6553, ENATS 12-8		
			4.8.1	Full load current -temperature rise		
			4.8.1	Carry permitted overload without deterioration		
			4.8.1	No impairment of safe breaking performance		
			4.8.1	Identification particulars		
			4.8.2	Rated normal current		
9	Information to be Given with enquiries, tenders and orders		4.9	Information to be Given with enquiries, tenders and orders- schedule 4.1		
10	Rules for transport, storage, erection, operation and maintenance		4.10	Rules for transport, storage, erection, operation and maintenance		
11	Safety		4.11	Safety		

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APPENDIX 5G ENATS 41-36 ANNEX D4 Metal Enclosed Switch-Fuse Combinations

SELF CERTIFICATION CONFORMANCE DECLARATION

PART 4 – METAL ENCLOSED SWITCH-FUSE COMBINATIONS - TYPE TEST CONFORMANCE DECLARATION D4

Type tests for switch-fuse including enclosure and earthing switch.

Manufacturer:

Product reference:

Name:

Signature:

Ratings

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
 - ENA/SAP to complete columns 1 to 4
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the table

Type test reports table based on ENATS 41-36 Table 4.1

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conformance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
1	Dielectric switch- earthing disconnecter and switch as appropriate. Partial discharge ≤ 10pC Type tests to be representative of two cable termination systems, in addition to switchgear manufacturer's own system if any	IEC 60694. Sub-clause 6.2, IEC 62271-105 . Sub-clause 6.2, IEC 62271-200. Sub-clause 6.2. Tables 1.1a and 1.1b of ENATS 41-36. IEC 62271-200. Sub-clause 6.2.9 and Annex B ENATS 41-18 Sub-clauses 1.5.103.1 and 1.6 of ENATS 41-36								
2	" - Busbars	"								
3	Voltage Withstand - Isolating Gap (provision for dielectric tests on cables).#	IEC 62271-200. Sub-clause 5.105 and 6.2.101. Sub-clause 1.4.2.2 of ENATS 41-36.								

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1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conform ance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
4	DC Withstand Test on Test Devices, including all parts of main circuit which cannot be disconnected from the test connections	IEC 62271-200. Sub-clause 5.105. Sub-clause 1.5.201.1 of ENATS 41-36.							
5	Insulation level - electrically stressed gap due to possible movement of earthing switch contacts	Sub-clause 1.5.6 of ENATS 41-36							
6	Measurement of the resistance of main circuit - Panel	IEC 60694. Sub-clause 6.4, IEC 62271-105 . Sub-clause 6.4, IEC 62271-200. Sub-clause 6.4							
7	Busbars								
8	Temperature Rise - Panel	IEC 60694. Sub-clause 6.5, IEC 62271-105 . Sub-clause 6.5, IEC 62271-200. Sub-clause 6.5							
9	- Busbars								
10	Short-time withstand current and peak withstand current tests – Earthing switch.(3sec short time) Incoming Outgoing	IEC 60694.Sub-clause 6.6, IEC 62271-105 . Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.							
11	- Busbars								
12	-Single phase test of earth circuit	IEC 60694.Sub-clause 6.6, IEC 62271-102. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.							
13	Verification of protection. (indoor – IP3X min) Weatherproofing for outdoor equipment. (IP3XDW min) Mechanical impact (indoor 2J, outdoor 5J)	IEC 60694.Sub-clause 6.7, IEC 62271-105 . Sub-clause 6.7, IEC 62271-200. Sub-clause 6.7 IEC 529 Sub-clause 1.5.13 of ENATS 41-36.							
14	Tightness test	IEC 60694.Sub-clause 6.8, IEC 62271-105 . Sub-clause 6.8, IEC 62271-200. Sub-clause 6.8.							
15	EMC tests	IEC 60694.Sub-clause 6.9, IEC 62271-105 . Sub-clause 6.9,							
16	Short-circuit making and breaking tests - Switch-fuse	IEC 62271-105 . Sub-clauses 6.101 IEC 62271-200. Sub-clause 6.101							
17	Short-circuit making tests - Earthing switch (class E2) Test duty 5 of IEC 60265 sub-clause 6.101.10 - 5 making operations. Incoming Outgoing	IEC 62271-102. Sub-clause 6.101, IEC 60265. Sub-clause 6.101.10							
18	Mechanical operations - Switch-fuse	IEC 62271-105 . Sub-clause 6.102., IEC 62271-200. Sub-clause 6.102							

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	1. Test Requirement	2. Specification and standards	3. Rated value	4. Test req'd Y or N	5. Conformance *	6. Test value	7. Date of test	8. Test station Report / Cert No	9. Witness I, M or ENA **	10. Remarks
19	Mechanical shock test on fuses	IEC 62271-105 . Sub-clause 6.103								
20	Mechanical operations - Earthing switch (including mechanical strength of kinematic chain between movable contacts and the position indicating device). Incoming Outgoing	IEC 62271-102. Sub-clause 6.102 IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
21	Mechanical operations -Switching devices and removable parts- 50 ops -Interlocks-50 ops(mechanical and electro-mechanical)	IEC 62271-200. Sub-clause 6.102								
22	Low temperature tests	IEC 62271-102. Sub-clause 6.104								
23	High temperature tests- subject to design (see clause 1.6)	IEC 62271-102. Sub-clause 6.104, Clause 1.6 of ENATS 41-36								
24	Thermal test with long pre arcing time of fuse	IEC 62271-105. Sub-clause 6.104								
25	Internal Arc -- Switch-fuse chamber, BB Chamber, Cable box (36kV cable boxes using screened separable connectors – min 5kA.)	IEC 62271-200. Sub-clause 6.106. Sub-clause 1.5.101 of ENATS 41-36								
26	Gas-filled Compartment Pressure Withstand	IEC 62271-200. Sub-clause 6.103								
27	Voltage presence indicating system (VPIS)	IEC 61958. Clause 6								
28	Voltage detecting device (VDS)	IEC 61243-5. Clause 5								
29	Finish	Performance to ENATS 98-1								
30	Process Control	ISO 9001								

Switch-fuse equipments without test facilities shall be subjected to this type test. Test voltage shall be applied between the busbars and the circuit or transformer terminals of the equipment with the equipment prepared for normal service and in the "OFF" position.

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* Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The test conforms fully with the requirements of this clause
Cs2 = The test conforms partially with the requirements of this clause
Cs3 = The test does not conform to the requirements of this clause
Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests
Ct2 = Not fully independent witnessed tests
Ct3 = Self verified tests
Ct4 = Alternative tests / evidence offered
Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.
Ct6 = Not tested

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Appendix 5F ENATS 41-36 ANNEX D5 Metal Enclosed Ring Main Equipment SELF CERTIFICATION CONFORMANCE DECLARATION

PART 5 – METAL ENCLOSED RING MAIN EQUIPMENT

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 5, 'METAL ENCLOSED RING MAIN EQUIPMENT'

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 5 and the clauses of the aforementioned Standards relevant to metal enclosed ring main equipment. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Metal enclosed ring main equipment comprises combinations of switchgear from Parts 2 and 3, or Parts 3 and 4 of ENATS 41-36 and the self certification conformance declarations in ANNEX D2, D3, or D4 are to be completed, as appropriate, in addition to ANNEX D5

This conformance declaration is to be completed in addition to the ENATS 41-36 – ANNEX D1– Self Certification Conformance Declaration Part 1, 'Common Clauses'.

For associated protection, instrumentation and metering equipment ENATS 41-36 – ANNEX D10 - Self Certification Conformance Declaration Part 10 is to be completed.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The test conforms fully with the requirements of this clause
Cs2 = The test conforms partially with the requirements of this clause
Cs3 = The test does not conform to the requirements of this clause
Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Ratings:

Date:

Document reference		NPS/003/014				
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ENATS 41-36 Part 5 Clause /Sub-clause	Requirement	Conformance codes	Remarks
5.1 “ “	General Combination (Parts 2 and 3) Combination (Parts 4 and 5)		
5.2 “	Service conditions Indoor and outdoor		
5.4.1	Rated operating sequence		
5.5 “ “ “ “ “ “	Design and construction Extensible / Non-extensible Mimic diagram Test points Symbols/colours - Annex C Handles, padlocking, labels, indicators -access Bolting to transformer		
5.6	Type tests - Table 5.1 or Table 5.2		
5.7	Routine tests		
5.8	Guide to the selection of ring main equipment		
5.9	Information to be given with enquiries, tenders and orders - schedule 5.1		
5.10	Rules for transport, storage, erection, operation and maintenance		
5.11	Safety		

Document reference		NPS/003/014							
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Appendix 5H ENATS 41-36 ANNEX D5 Metal Enclosed Ring Main Equipment SELF CERTIFICATION CONFORMANCE DECLARATION

PART 5 – METAL ENCLOSED RING MAIN EQUIPMENT - TYPE TEST CONFORMANCE DECLARATION 5.2

Type tests for ring main equipment (RME) with circuit-breaker tee-off .

Manufacturer:

Ratings

Product reference:

Name:

Signature:

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
 - ENA/SAP to complete columns 1 to 4
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the

Type test reports table based on ENATS 41-36 Table 5.2

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specifications and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report/Cert No	Witness I, M or ENA**	Remarks
1	Dielectric - RME, including switch-disconnectors, circuit-breaker and earthing switches as appropriate. Tests to include VT and CTs when fitted. Partial discharge. Tests to be representative of two cable termination systems, in addition to switchgear manufacturer's own system if any.	IEC 60694. Sub-clause 6.2, IEC 62271-100. Sub-clause 6.2, IEC 60265: Part 1. Sub-clause 6.2, IEC 62271-200. Sub-clause 6.2. Tables 1.1a and 1.1b of ENATS 41-36 IEC 62271-200. Sub-clause 6.2.9 and Annex B ENATS 41-18 Sub-clauses 1.5.103.1 and 1.6 of ENATS 41-36								
2	voltage Withstand – Isolating Gap (provision for dielectric tests on cables).	IEC62271-200. Sub-clauses 5.105 and 6.2.101. Sub-clause 1.4.2.2 of ENATS 41-36.								

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1.	Test Requirement	2. Specifications and Standards	3. Rated value	4. Test req'd Y or N	5. Conformance	6. Test value	7. Date of test	8. Test station Report/Cert No	9. Witness I, M or ENA**	10. Remarks
3	d.c. Withstand Test on Test Devices, including all parts of the main circuit which cannot be disconnected from the test connections.	IEC 62271-200. Sub-clause 5.105. Sub-clause 1.5.201.1 of ENATS 41-36								
4	Insulation level - electrically stressed gap due to movement of earthing switch contacts.	Sub-clause 1.5.6 of ENATS 41-36								
5	Measurement of the resistance of main circuit - RME, including switch -disconnectors and circuit-breaker as appropriate.	IEC 60694. Sub-clause 6.4, IEC 62271-100. Sub-clause 6.4. IEC 60265: Part 1. Sub-clause 6.4, IEC 62271-200. Sub-clause 6.4								
6	Temperature Rise - RME, including switch - disconnectors and circuit breaker as appropriate.	IEC 60694. Sub-clause 6.5, IEC 62271-100. Sub-clause 6.5 IEC 60265: Part 1. Sub-clause 6.5, IEC 62271-200. Sub-clause 6.5								
7	Short-time withstand current and peak withstand current tests - Ring switch-disconnectors, busbar and connections .(3sec short time).	IEC 60694. Sub-clause 6.6, IEC 60265: PART 1. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
8	Short-time withstand current and peak withstand current tests - Tee-off circuit-breaker. (3sec short time).	IEC 60694. Sub-clause 6.6, IEC 62271-100. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
9	Short-time withstand current and peak withstand current tests - Tee-off Earthing switch (3sec short time)..	IEC 60694. Sub-clause 6.6, IEC 62271-102. Sub-clause 6.6, IEC 62271-200. Sub-clause 6.6.								
10	" - Ring Earthing switches.	"								
11	" - Single phase test of earth circuit. (sub-clause 1.4.5 of ENATS 41-36)	IEC60694. Sub-clause 6.6, IEC62271-100. Sub-clause 6.6, IEC62271-200. Sub-clause 6.6.								
12	Verification of protection. (indoor – IP3X minimum) Weatherproofing for outdoor equipment. (outdoor IP3XDW minimum) Mechanical impact. (indoor – 2J, outdoor – 5J)	IEC 60694. Sub-clause 6.7 and annex C IEC 529, IEC 62271-200. Sub-clause 6.7 Sub-clause 1.5.13 of ENATS 41-36.								
13	Tightness test.	IEC 60694. Sub-clause 6.8, IEC 62271-200. Sub-clause 6.8.								
14	EMC tests.	IEC 60694. Sub-clause 6.9								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specifications and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report/Cert No	Witness I, M or ENA**	Remarks
15	Short-circuit making and breaking tests - Circuit-breaker class E2. Non auto-reclosing circuit-breaker - tested in accordance with sub-clauses 6.112.1 and 6.106 of IEC 62271-100.	IEC62271-100. Sub-clauses 6.102 to 6.106, and 6.112 IEC62271-200. Sub-clause 6.101								
16	Short-circuit making and breaking tests - Ring switch-disconnector (Switch-disconnector class E3 and Test duty 5 = 10 operations).	IEC 60265: PART 1.Sub-clauses 6.101, (TD1 to 5, Table 5), IEC 62271-200. Sub-clause 6.101								
17	Short-circuit making tests - Tee-off earthing switch (class E2) (Test duty 5 of IEC 60265 sub-clause 6.101.10 - 5 making operations)	IEC 62271-102. Sub-clause 6.101, IEC60265. Sub-clause 6.101.10								
18	Short-circuit making tests - Ring earth switch (class E2) (Test duty 5 of IEC 60265 sub-clause 6.101.10 - 5 making operations).	IEC 62271-102. Sub-clause 6.101, IEC60265. Sub-clause 6.101.10								
19	Mechanical operations - Circuit-breaker. Non auto-reclosing circuit-breaker - class M1 - 2000 operating cycles. If used as earthing device - mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC62271-100. Sub-clause 6.101.2. IEC62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
20	Mechanical operations - Ring switch-disconnector - class M2 General purpose switch - 5,000 operations. (Including mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC 60265: PART 1. Sub-clause 6.102.4, IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
21	Mechanical operations - Tee-off earthing switch. (Including mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC 62271-102. Sub-clause 6.102 IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specifications and Standards	Rated value	Test req'd Y or N	Conform ance	Test value	Date of test	Test station Report/Ce rt No	Witness I, M or ENA**	Remarks
22	Mechanical operations - Ring earthing switches. (Including mechanical strength of kinematic chain between movable contacts and the position indicating device).	IEC 62271-102. Sub-clause 6.102 IEC 62271-200. Sub-clause 6.102 IEC 62271-102. Sub-clause 6.105 and Annex A								
23	Mechanical operations- switching devices and removable parts - 50 ops - interlocks - 50 ops(mechanical and electro-mechanical).	IEC 62271-200. Sub-clause 6.102								
24	Low temperature tests.	IEC 62271-100. Sub-clause 6.101.3								
25	High temperature tests - subject to design (see clause 1.6 of this specification).	IEC 62271-100.Sub-clause 6.101.3								
26	Out of phase breaking current – Circuit-breaker	IEC62271-100. Sub-clause 6.110								
27	Cable-charging breaking current tests - Ring switch- disconnectors.	IEC 60265: PART 1. Sub-clause 6.101.8.4(f)								
28	Cable-charging breaking current tests - Circuit-breaker.	IEC 62271-100. Sub-clause6.111								
29	Line-charging breaking current tests - Ring switch- disconnectors.	IEC 60265: PART 1. Sub-clause 6.101.8.4(g)								
30	Line-charging breaking current tests - Circuit-breaker.	IEC 62271-100, Sub-clause 6.111								
31	Internal Arc –C B Chamber, C T Chamber, BB Chamber, Cable box. (36kV cable-boxes using screened separate connectors – min 5kA)	IEC62271-200. Sub-clause 6.106 and Annex A. Sub-clause 1.5.101 of ENATS 41-36								
32	Gas-filled Compartment Pressure Withstand.	IEC 62271-200. Sub-clause 6.103								
33	Voltage presence indicating system (VPIS).	IEC 61958. Clause 6								
34	Voltage detecting system (VDS)	IEC 61243-5. Clause 5								
35	Finish.	Performance to ENATS 98-1								
36	Process Control.	ISO 9001								

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* Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests

Ct2 = Not fully independent witnessed tests

Ct3 = Self verified tests

Ct4 = Alternative tests / evidence offered

Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.

Ct6 = Not tested

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Appendix 5I ENATS 41-36 ANNEX D5 Protection, Instrumentation and Metering Equipment

PART 10 – PROTECTION, INSTRUMENTATION AND METERING EQUIPMENT

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36 – Part 10, ‘PROTECTION, INSTRUMENTATION AND METERING EQUIPMENT’

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 10 and the clauses of the aforementioned Standards relevant to protection, instrumentation and metering equipment.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.
Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The product conforms fully with the requirements of this clause
Cs2 = The product conforms partially with the requirements of this clause
Cs3 = The product does not conform to the requirements of this clause
Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant ‘IEC’ or ‘ENATS’ as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Ratings:

Date:

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ENATS 41-36 Part 10 Clause / Sub- clause	Requirements	Conformance code	Remarks
10.1	General		
10.1	Ratings as associated switching device		
10.1	ENATS 35-15		
10.2	Current transformers		
10.2.1	General		
10.2.1	CTs to IEC 60044 Part 1		
10.2.1	Characteristics to ENATS 35-15		
10.2.1	Position of CT data plates		
10.2.1	Secondary winding connections		
10.2.1	No common leads (metering CTs)		
10.2.1	Earth screen – accessible		
10.2.1	ER S15 and ENATS 50-18		
10.2.1	Individual test certificates		
10.2.1	Low energy output devices		
10.2.1	CT installation		
10.2.2	Performance characteristics ENATS 35-15		
10.3	Voltage transformers		
10.3.1	General		
10.3.1	VTs to IEC 60044-2		
10.3.1	Dry, encapsulated, isolatable		
10.3.1	VT design		
10.3.1	Low energy output devices		
10.3.1	Individual test certificates		
10.3.1	Prevent access to metering circuits		
10.3.2	Performance characteristics ENATS 35-15		
10.3.3	VT connections		
10.3.3.1	General		
10.3.3.1	Means of breaking primary connections		
10.3.3.1	Secondary windings fuses and links		
10.3.3.1	Removal of secondary fuse-links in service		
10.3.3.1	Means of breaking connections labelled		
10.3.3.1	Padlockable shutters – primary isolation		
10.3.3.1	Primary connection fuse links – BS 2692		
10.3.3.1	Fuse rating – 3.15A		
10.3.3.1	Fuses for oil insulated VT		
10.3.3.2	Star point connections compartment – fixed cover		
10.3.3.2	Fixed cover labelled		
10.3.4	Padlocking facilities		

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ENATS 41-36 Part 10 Clause / Sub- clause	Requirements	Conformance code	Remarks
10.3.4.1	Safety padlocking facilities a) to c)		
10.4	Metering equipment		
10.4	LV connections brought out		
10.4	Sealable terminal block		
10.4	VT accuracy class 1.0 (10MVA)		
10.4	VT accuracy class 0.5 (>10MVA)		
10.4	Two windings as (a) or (b)		
10.4	CT accuracy class 0.5S (10MVA)		
10.4	CT accuracy class 0.2S (<100MVA)		
10.4.1	Self contained metering unit		
10.4.1	Metal enclosed free standing		
10.4.1	Flange for ENATS 35-1 transformer		
10.4.1	Multicore cable box – BS6121 (E1W)		
10.5	Instruments		
10.5	ENATS 50-18		
10.5	Scale		
10.6.1	Earth fault passage indication		
10.6.1	Readily visible		
10.6.1	Core balance CT		
10.6.1	Insulated cable gland		
10.6.2	Remote earth fault indication		

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Appendix 5J ENATS 48-05

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 48-5 – ‘Environmental Test Requirements for protection Relays and Systems’

Protection relays shall comply with the latest issues of the relevant International and British Standards. ENATS 48-05 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 48-05 relevant to Environmental Test Requirements for protection relays and systems.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.
Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The product conforms fully with the requirements of this clause
Cs2 = The product conforms partially with the requirements of this clause
Cs3 = The product does not conform to the requirements of this clause
Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant ‘IEC’ or ‘ENATS’ as appropriate

Manufacturer:
Relay Reference :
Name:

Signature:

Relay Type:

Date:

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Protection Relay Type Testing – Atmospheric Environment Requirements

Standard	Test	Requirement	Conformance Code	Remarks
IEC 60068-2-1	Temperature Cold Heat	-25°C, 96 hours, operate and storage		
IEC 60068-2-2	Temperature Dry Heat	-70°C, 96 hours, operate and storage		
IEC 60068-2-3	Relative Humidity			
IEC 60068-2-30	Relative Humidity Alternative	100% RH, 40°C, 56 cycles of +25°C to +55°C		
IEC 60529	Enclosure	IP54		

Protection Relay Type Testing Continued – Mechanical Environment Requirements

Standard	Test	Requirement	Conformance Code	Remarks
IEC 60255-21-1	Vibration	Response and Endurance Class 1		
IEC 60255-21-2	Shock	Response and Endurance Class 1		
IEC 60255-21-2	Bump	Class 1		
IEC 60255-21-3	Seismic	Class 1		

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Protection Relay Type Testing Continued – Electrical Requirements				
Standard	Test	Requirement	Conformance Code	Remarks
IEC 60255-6	48 V DC Supply Voltage	Table 1, 38.5 to 53 V with a maximum of > 60 V		
IEC 60255-11	Voltage Dips, Short Interruptions and Voltage Immunity Test	10ms interruption, 12% AC ripple		
	DC Supply Voltage General	Ramp up and down over 1 minute or similar		
ENA TS 48-4	Low Burden Trip Relays	Captive discharge ENA 1		
ENA TS 48-4	High Burden Trip Relays	Captive discharge ENA 1		
ENA TS 48 – 5	Thermal requirement of CT inputs	2.4 x In, continuous 3.0A, 20 mins 3.5A, 10 mins 4.0A, 5 mins 5.0A, 3 mins 6.0A, 2 mins		
ENA TS 48 – 5	Thermal requirements of VT inputs	120% of Vn, continuous		
ENA TS 48 - 5	Dielectric	Test values selected according to insulation voltage. High Impedance circulating current schemes, test at 2.5kV. Circuits connected to instrument transformers or batteries, rated insulation not below 250V, test at 2.0kV. Open output relay contacts 1kV.		
IEC 60255 - 5	Impulse Voltage	Test at 5kV, 0.5J		

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Electromagnetic Compatibility (EMC) Requirements					
Standard	Test	Requirement	Conformance Code	Remarks	
IEC 60255	Oscillatory waves immunity test (High Frequency Disturbance)	Class III, 1MHz 2.5kV common, 1 kV diff. Applied to all ports, except diff on comms at the discretion of the approving panel			
IEC 60255	Electrostatic Discharge Immunity Tests	Class III. 6kV, contact, 8kV air. Applied to the enclosure			
IEC 60255	Radiated electromagnetic Field Disturbance test (RFI)	10V/m, 1kHz, 80 to 1000MHz sweep and 80, 160, 450, 900 Mhz spot frequencies. Applied to the enclosure			
IEC 60255	Radiated electromagnetic field from digital radio telephones immunity test	10V/m, 900 and 1890Mhz. Applied to the enclosure			
IEC 60255	Electrical fast transient/burst immunity	Level IV, 4kV. Applied to all ports			
	Surge immunity	Level III, 2kV common, 1kV diff. (Level 4, 4kV, 2kV preferred for CT and VT inputs) Applied to all ports			
IEC 60255	Conducted electromagnetic field disturbance tests	10Vrms, 80% mod, 1kHz. 0.15 to 80MHz sweep and 27 and 68 MHz spot frequencies. Applied to all ports			
EN 61000-4-8 (IEC 61000-4-8)	Power frequency magnetic field immunity test	1000 A/m for 1 sec and 100 A/m for 1 min. Applied to all ports			
EN 61000-4-9 (IEC 61000-4-9)	Pulse magnetic field immunity test	6.4/16 μ s magnetic pulse, 1000 A/m. Applied to enclosure. Not currently mandatory			
EN 61000-4-10 (IEC 61000-4-10)	Damped oscillatory magnetic field immunity test	0.1 and 1.0 MHz, 100A/m. Applied to the enclosure. Not currently mandatory			
IEC 60255-22-7 & EATS 48-5	Power Frequency	Level 4, 300V for 1 s at 50Hz, common mode. Appendix A, EATS 48-5 for differential, at discretion of panel			
IEC 60834-1 & IEC 60834-2	Communication channel Noise immunity				
IEC 60255-25	Conducted and radiated Emission	Class A, conducted, power supply: 0.15 to 0.5 MHz, 79dB(μ V) quasi peak, 66dB (μ V) average, 0.5 to 30 MHz, 71dB(μ V) quasi peak, 60dB (μ V) average. Radiated, enclosure at 10M: 30 to 230 MHz, 40dB(μ V) quasi peak, 230 to 1000 MHz, 47dB(μ V) quasi peak,			

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Appendix 5K ENATS 48-6-6

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 48-6-6 'Functional Test Requirements – Overcurrent and Earth Fault Protection

Protection relays shall comply with the latest issues of ENATS 48-6-6. The functional tests detailed below are intended to show compliance with the manufacturers 'claimed operating characteristics and performance specifications and to demonstrate correct performance during fault scenarios

This check sheet identifies the clauses in ENATS 48-6-6 relevant to Overcurrent and Earth Fault Protection systems.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.
Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
Cs1 = The product conforms fully with the requirements of this clause
Cs2 = The product conforms partially with the requirements of this clause
Cs3 = The product does not conform to the requirements of this clause
Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:
Relay Reference :
Name:

Signature:

Relay Type:

Date:

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ENATS 48-6-6 clause / Sub- clause	Requirements	Conformance code	Remarks
10.3.4.1	Safety padlocking facilities a) to c)		
10.4	Metering equipment		
10.4	LV connections brought out		
10.4	Sealable terminal block		
10.4	VT accuracy class 1.0 (10MVA)		
10.4	VT accuracy class 0.5 (>10MVA)		
10.4	Two windings as (a) or (b)		
10.4	CT accuracy class 0.5S (10MVA)		
10.4	CT accuracy class 0.2S (<100MVA)		
10.4.1	Self contained metering unit		
10.4.1	Metal enclosed free standing		
10.4.1	Flange for ENATS 35-1 transformer		
10.4.1	Multicore cable box – BS6121 (E1W)		
10.5	Instruments		
10.5	ENATS 50-18		
10.5	Scale		
10.6.1	Earth fault passage indication		
10.6.1	Readily visible		
10.6.1	Core balance CT		
10.6.1	Insulated cable gland		
10.6.2	Remote earth fault indication		

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Appendix 6 – Addendum to Suppliers Requirements

Appendix 3, 4 & 5 contains self certification conformance declarations. They shall be completed by suppliers tendering products for this contract with one set completed for each variant offered. Type test evidence and/or samples shall be submitted when requested.

Suppliers shall confirm their ability to carry out the range or pre-commissioning checks detailed in Appendix 2 “Testing”

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Appendix 7 – Summary major document changes

Clause No.	Summary of change to clause
3.1.3	Remote Control – this clause has been expanded to provide additional detail including reference to NPS/003/017 - Technical Specification for a Ground Mounted Distribution Switchgear Remote Control System.
3.3	Protection – this clause has been expanded to include additional detail on mid point CB protection requirements
3.3.1	Relays – This clause has been amended to detail the need for relays to have been ENA assessed or have been self certified in accordance with Appendix 5J & K
3.4	Testing – new clause detailing the required test voltages that are applied to cables terminated in the switchgear. Additionally this clause includes details about VT and CT connections.
3.5	Resistance to Physical Interference/Damage – this clause has been amended to include the need for metallic screens or covers over separable connectors on cable terminations
3.6	Variances from ENATS 41-36 -
3.6 / 1.4.4.1	Rated short time withstand current increased at 24kV from 12kA to 16kA min
3.6 / 1.4.8	Rated supply voltage specified for remote control and emergency trip
3.6 / 1.5.13.2	Clarification of IP rating for protection against water ingress for indoor switchgear
3.6 / 10.6.1	Added reference to NPS/001/013 - Fault Passage Indicators.
4.1	References added to ENATS and IEC specifications to reflect changes above
4.2	References added to CE Electric UK CDS NPS specifications to reflect changes above
6.2	Policy Sponsor changed from Geoff Bolden to Mark Nicholson
Appendices	Appendix numbering rearranged to accommodate the insertion of test voltages table in Appendix 2
Appendix 1	Metering interface terminal positioning updated to avoid the need to enter compartment with primary conductors, to reflect operational experiences Added typical requirements for site that exports onto CE Electric UK network
Appendix 2	Added Table of CE Electric UK OPM test voltages
Appendix 4	Technical Details Declaration table updated to include critical gas pressures
Appendix 6	Added requirement for manufacturers to carry out pre-commissioning tests and supply CE Electric UK with associated test certificates.
Appendix 7	Added to document material changes to this specification since the last issue.